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Part I

Cyclic analyses

1 crushing compression — solved tension tail — perfect bond

1.1 Cyclic — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — ladder to $\pm 0.004\%$

2026-09-05_191849_cyclic_c02-solved_nobond_d4e-05_protoladder

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_rupture, concrete_residual, steel_b — this run pre-dates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, $L = 50 \text{ mm}$
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, $L = 70.71 \text{ mm}$
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, $L = 50 \text{ mm}$

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	$L \text{ (mm)}$	$A \text{ (mm}^2\text{)}$	$E \text{ (MPa)}$	$EA \text{ (N)}$	$EA/L \text{ (N/mm)}$
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	cyclic
protocol	ladder, 1 cycle(s) per level, largest $\pm 0.004\%$ drift
levels (% drift)	0.0002, 0.0004, 0.0006, 0.0008, 0.0012, 0.002, 0.003, 0.004
drive path	1.1 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional

steps	18,060
wall clock	0.03 h
converged	True

Recorded response

peak base shear	134.9 kN (1 ms moving average)
raw sample maximum	156.5 kN
drift at peak	0.004%
drift capacity (20% drop)	not reached — basis: cyclic tip envelope, 16 tips over 15 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	35.6 kN

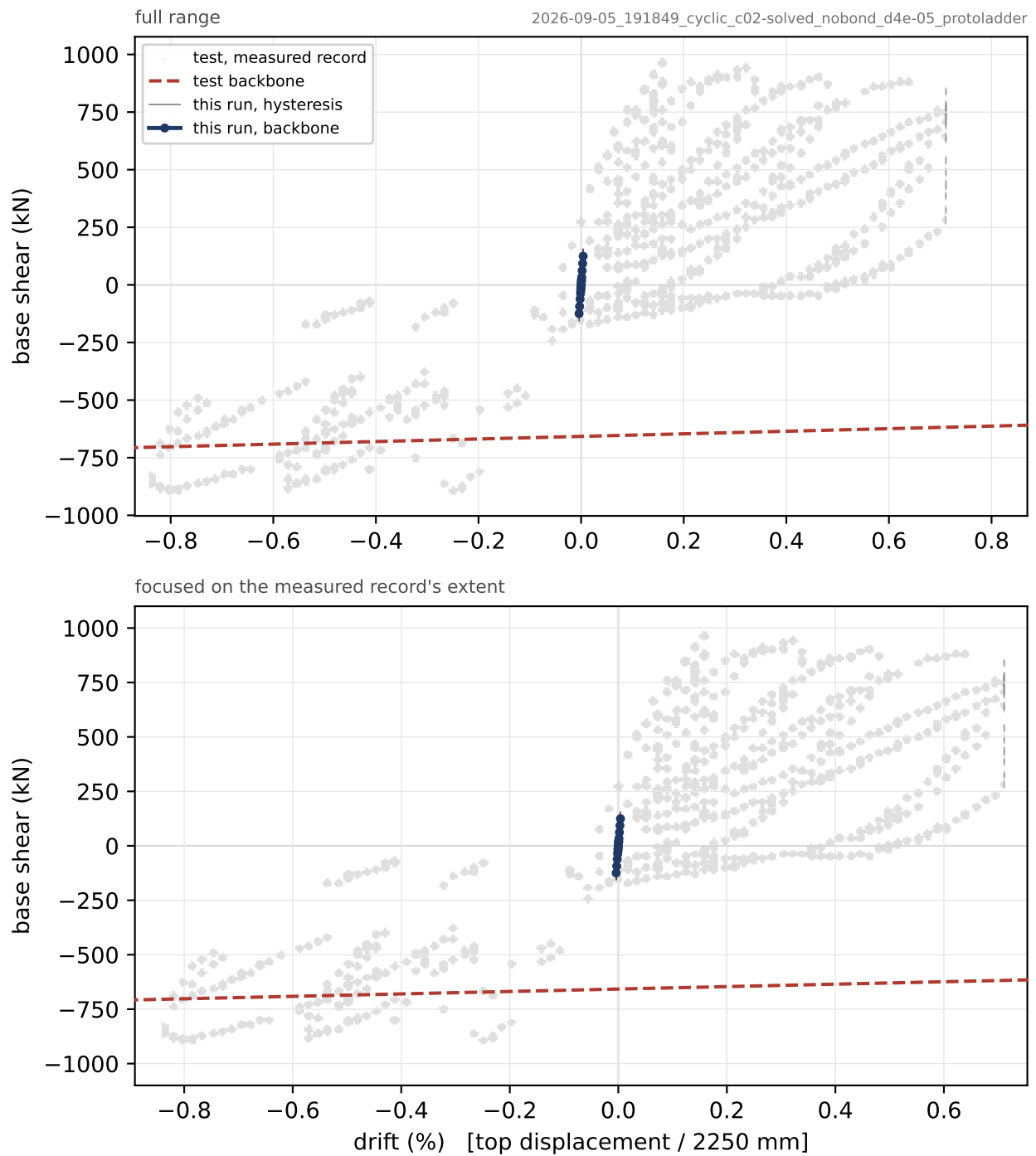


Figure 1: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

1.2 Cyclic — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — ladder to $\pm 1\%$

2026-09-06_015255_cyclic_c02-solved_nobond_d1pct_protoladder

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	Concrete02	<code>fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83</code>	5,505 concrete elements, $L = 50$ mm
2	Concrete02	<code>fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38</code>	5,400 concrete elements, $L = 70.71$ mm
3	Steel02	<code>Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15</code>	2,626 longitudinal/stirrup elements, $L = 50$ mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	cyclic
protocol	ladder, 1 cycle(s) per level, largest $\pm 1\%$ drift
levels (% drift)	0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.75, 1
drive path	274.5 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	4,514,700
wall clock	7.75 h
converged	True

Recorded response

peak base shear	918.7 kN (1 ms moving average)
raw sample maximum	918.8 kN
drift at peak	0.5%
drift capacity (20% drop)	not reached — basis: cyclic tip envelope, 16 tips over 16 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	6.1 kN

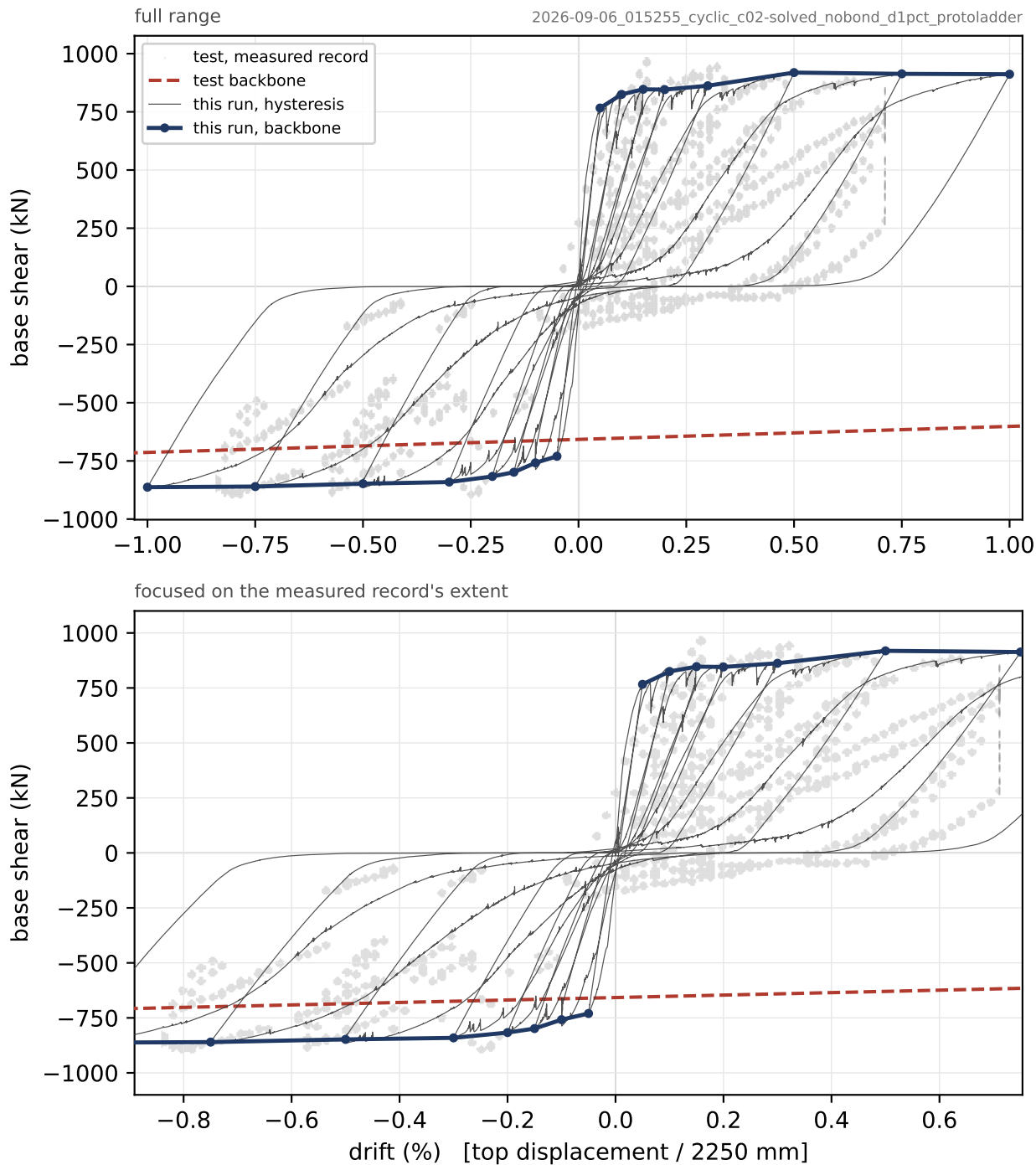


Figure 2: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

1.3 Cyclic — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — ext2p0 to $\pm 2\%$

2026-09-06_102244_cyclic_c02-solved_nobond_d2pct_protoext2p0

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	Concrete02	<code>fpc:-28</code> , <code>epsc0:-0.00225171</code> , <code>fpcu:-5.6</code> , <code>epsU:-0.0245731</code> , <code>lambda:0.1</code> , <code>ft:1.85</code> , <code>Ets:1140.83</code>	5,505 concrete elements, $L = 50$ mm
2	Concrete02	<code>fpc:-28</code> , <code>epsc0:-0.00225171</code> , <code>fpcu:-5.6</code> , <code>epsU:-0.0180353</code> , <code>lambda:0.1</code> , <code>ft:1.85</code> , <code>Ets:1613.38</code>	5,400 concrete elements, $L = 70.71$ mm
3	Steel02	<code>Fy:360</code> , <code>E0:200000</code> , <code>b:0.01</code> , <code>R0:18</code> , <code>cR1:0.925</code> , <code>cR2:0.15</code>	2,626 longitudinal/stirrup elements, $L = 50$ mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	cyclic
protocol	ext2p0, 1 cycle(s) per level, largest $\pm 2\%$ drift
levels (% drift)	0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2
drive path	859.5 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	14,136,189
wall clock	27.07 h
converged	True

Recorded response

peak base shear	918.7 kN (1 ms moving average)
raw sample maximum	918.8 kN
drift at peak	0.5%
drift capacity (20% drop)	not reached — basis: cyclic tip envelope, 22 tips over 22 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	6.1 kN

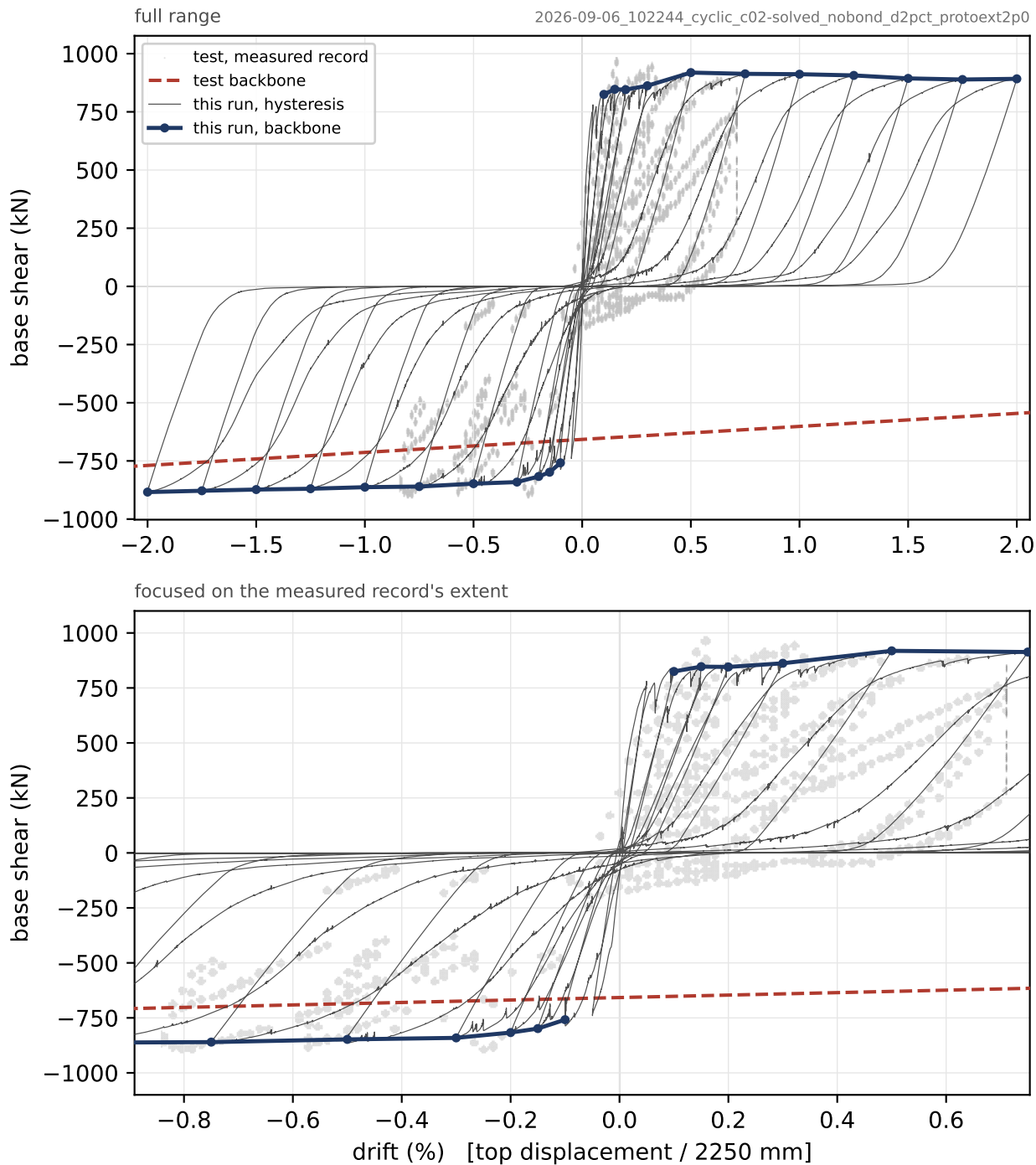


Figure 3: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

1.4 Cyclic — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $b = 0$ — ladder to $\pm 1.5\%$

2026-09-10_232041_cyclic_crushing-solved-perfect_eps0.05_res0_sb0_d1.5pct_protoladder

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	cyclic
protocol	ladder, 1 cycle(s) per level, largest $\pm 1.5\%$ drift
levels (% drift)	0.075, 0.15, 0.225, 0.3, 0.45, 0.75, 1.125, 1.5
drive path	411.8 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	6,772,049
wall clock	11.44 h
converged	True

Recorded response

peak base shear	853.8 kN (1 ms moving average)
raw sample maximum	853.8 kN
drift at peak	0.21%
drift capacity (20% drop)	0.503% — basis: cyclic tip envelope, 16 tips over 16 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	5.4 kN

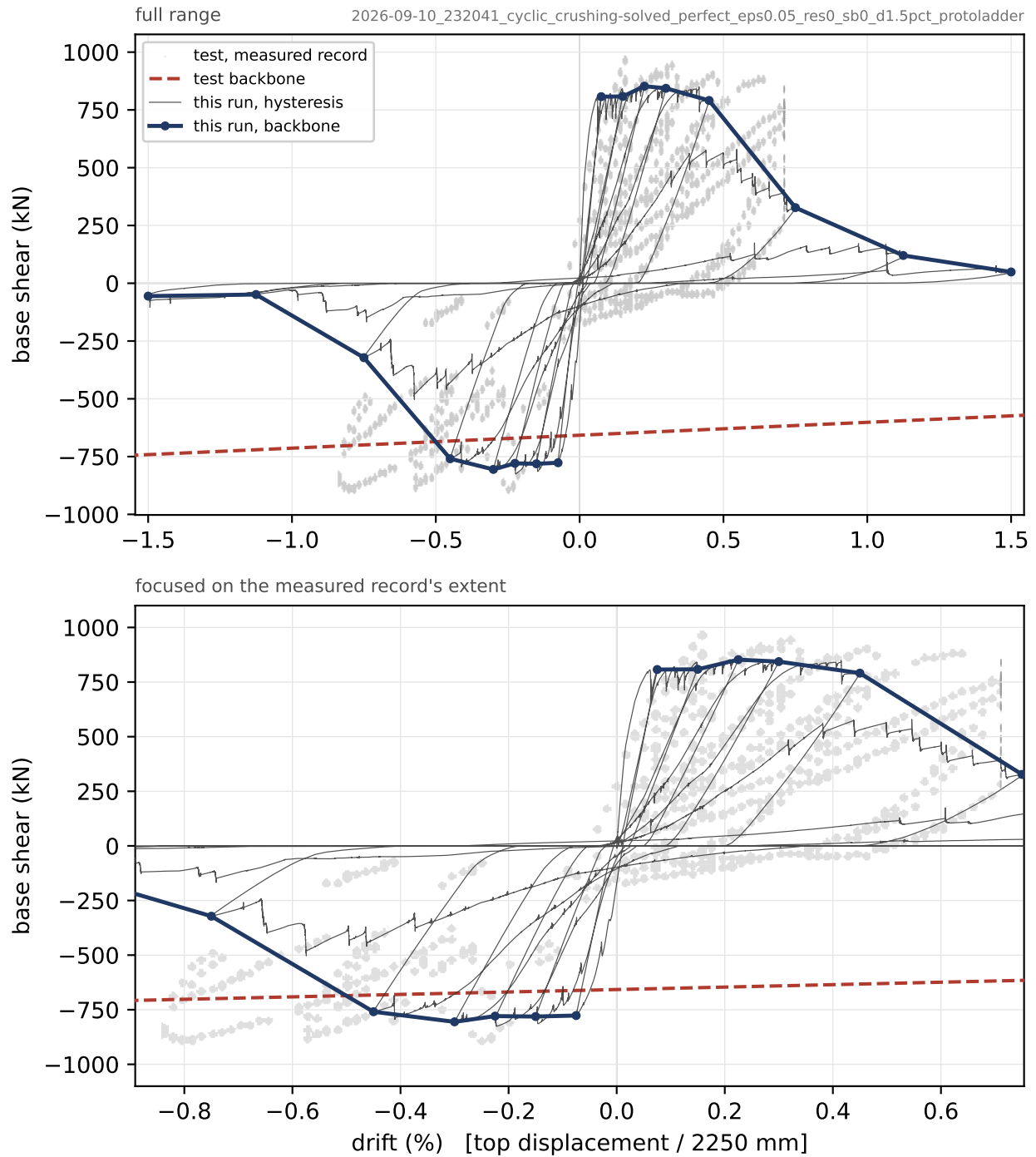


Figure 4: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

1.5 Cyclic — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — ladder to $\pm 1.5\%$

2026-09-10_074239_cyclic_crushing-solved_perfect_eps0.05_res0_d1.5pct_protoladder

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x G_f by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	cyclic
protocol	ladder, 1 cycle(s) per level, largest $\pm 1.5\%$ drift
levels (% drift)	0.075, 0.15, 0.225, 0.3, 0.45, 0.75, 1.125, 1.5
drive path	411.8 mm
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	6,772,049
wall clock	11.87 h
converged	True

Recorded response

peak base shear	931.1 kN (1 ms moving average)
raw sample maximum	931.3 kN
drift at peak	0.744%
drift capacity (20% drop)	not reached — basis: cyclic tip envelope, 16 tips over 16 levels
traced to	0% drift
inertia + damping in the reaction, ascending branch (p95)	5.9 kN

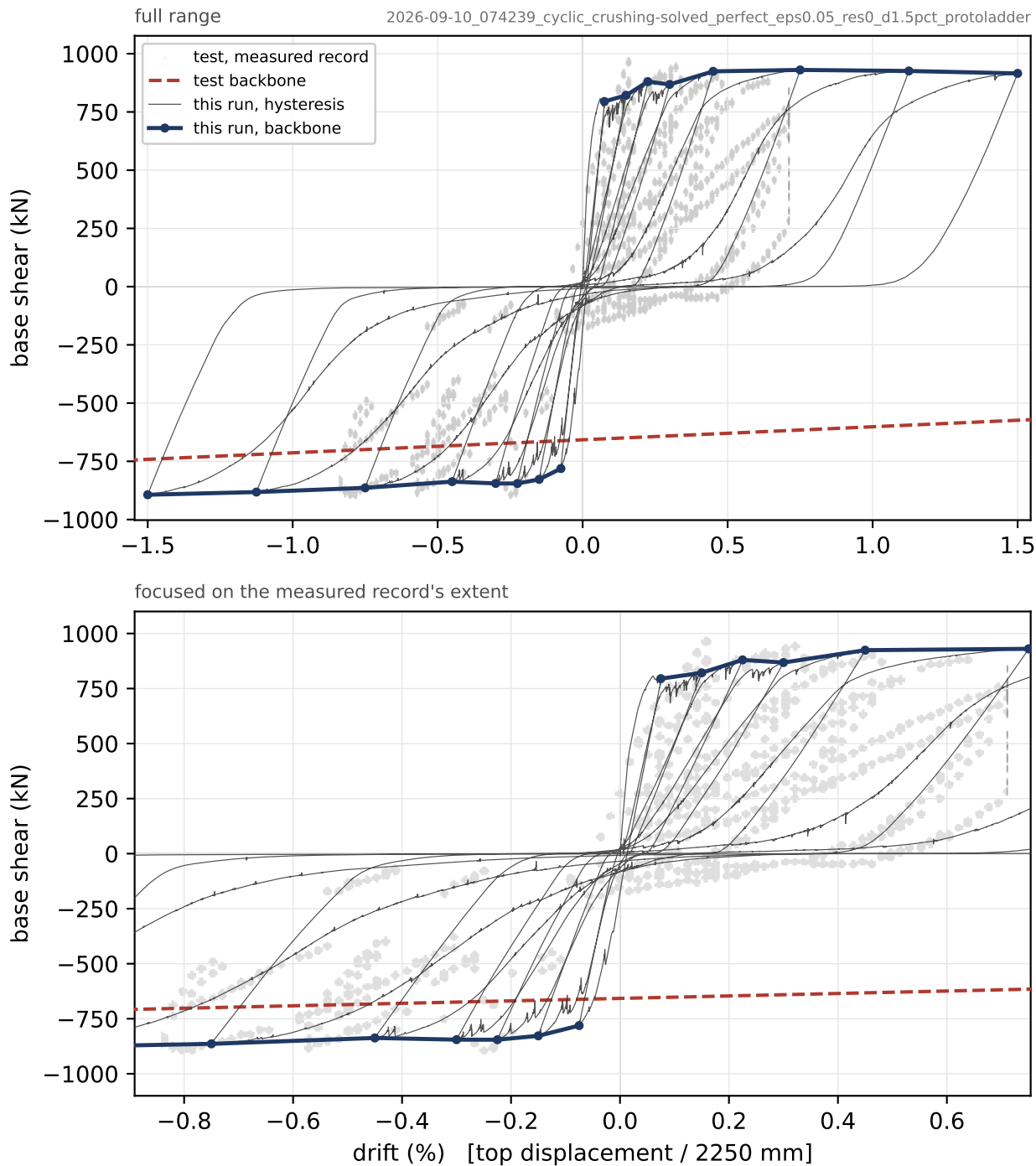


Figure 5: Hysteresis and its backbone over the measured record and its envelope; full range above, the record's own extent below.

Part II

Pushover analyses

2 capped compression — paper tension tail — perfect bond

2.1 Pushover — capped/paper/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_203422_pushover_eppcomp-paper_nobond

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	5.26 (5.26 x Gf per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	elastic-perfectly-plastic in compression, capped at f_c , no crushing
tension tail	the printed a_2, a_3 of Table 1, transplanted from the 20 mm grid
crushing residual	$0.2 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.00520708, 0.0267793, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, L = 50 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.00520708, 0.0267793, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	111,018

wall clock	0.21 h
converged	True

Recorded response

peak base shear	1,149.8 kN (1 ms moving average)
raw sample maximum	1,158 kN
drift at peak	0.3%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.3% drift
inertia + damping in the reaction,	7.2 kN
ascending branch (p95)	

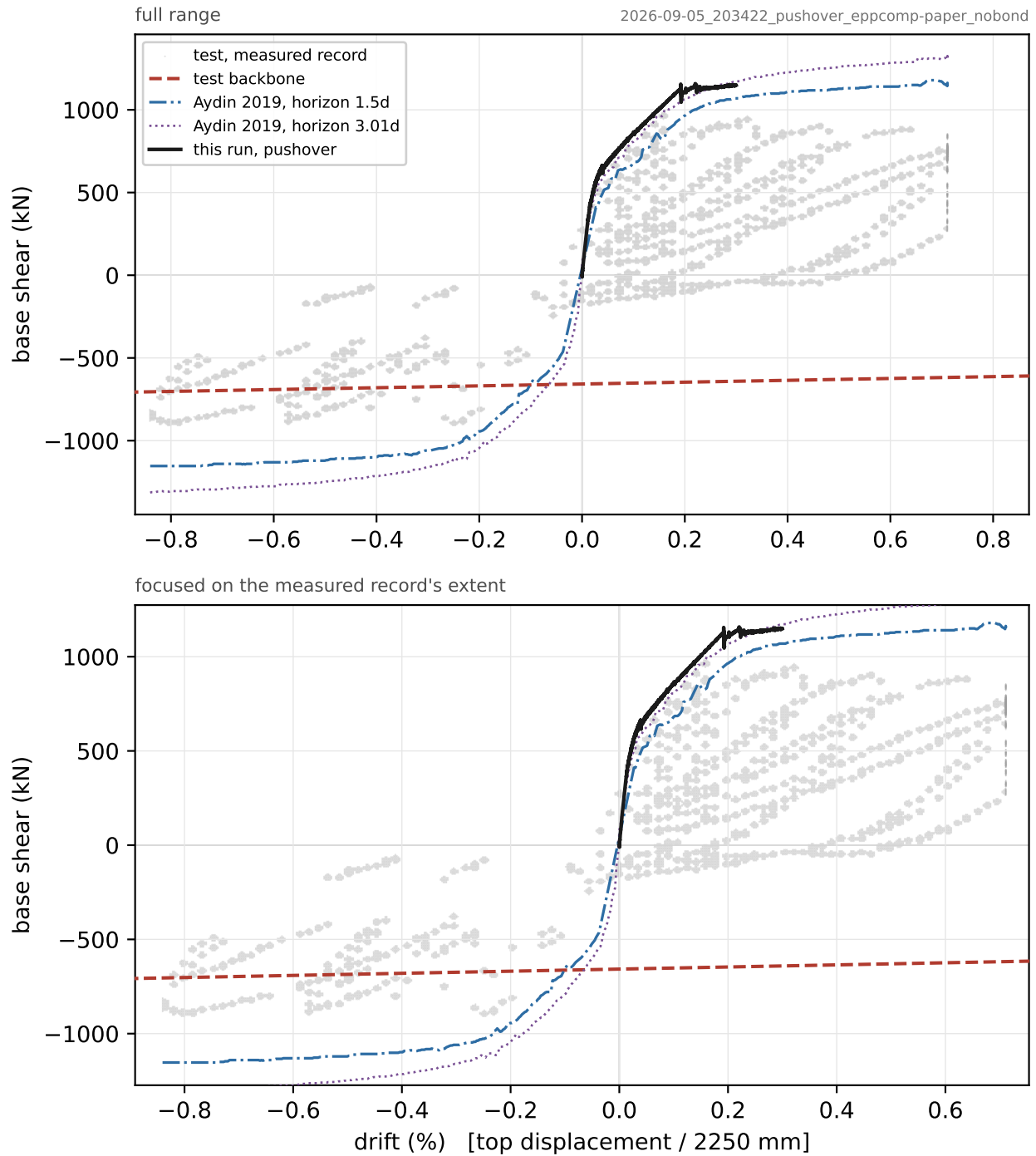


Figure 6: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3 capped compression — solved tension tail — perfect bond

3.1 Pushover — capped/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_202140_pushover_eppcomp-solved_nobond

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times$ mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7$ mm ²
$E_t A_t$ per strut	$157.3e6$ N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	elastic-perfectly-plastic in compression, capped at f_c , no crushing
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$

steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, $L = 50 \text{ mm}$
2	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, $L = 70.71 \text{ mm}$
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, $L = 50 \text{ mm}$

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	$L \text{ (mm)}$	$A \text{ (mm}^2\text{)}$	$E \text{ (MPa)}$	$EA \text{ (N)}$	$EA/L \text{ (N/mm)}$
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μs , 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	111,018
wall clock	0.21 h
converged	True

Recorded response

peak base shear	967.7 kN (1 ms moving average)
raw sample maximum	968.3 kN
drift at peak	0.3%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.3% drift
inertia + damping in the reaction, ascending branch (p95)	11.1 kN

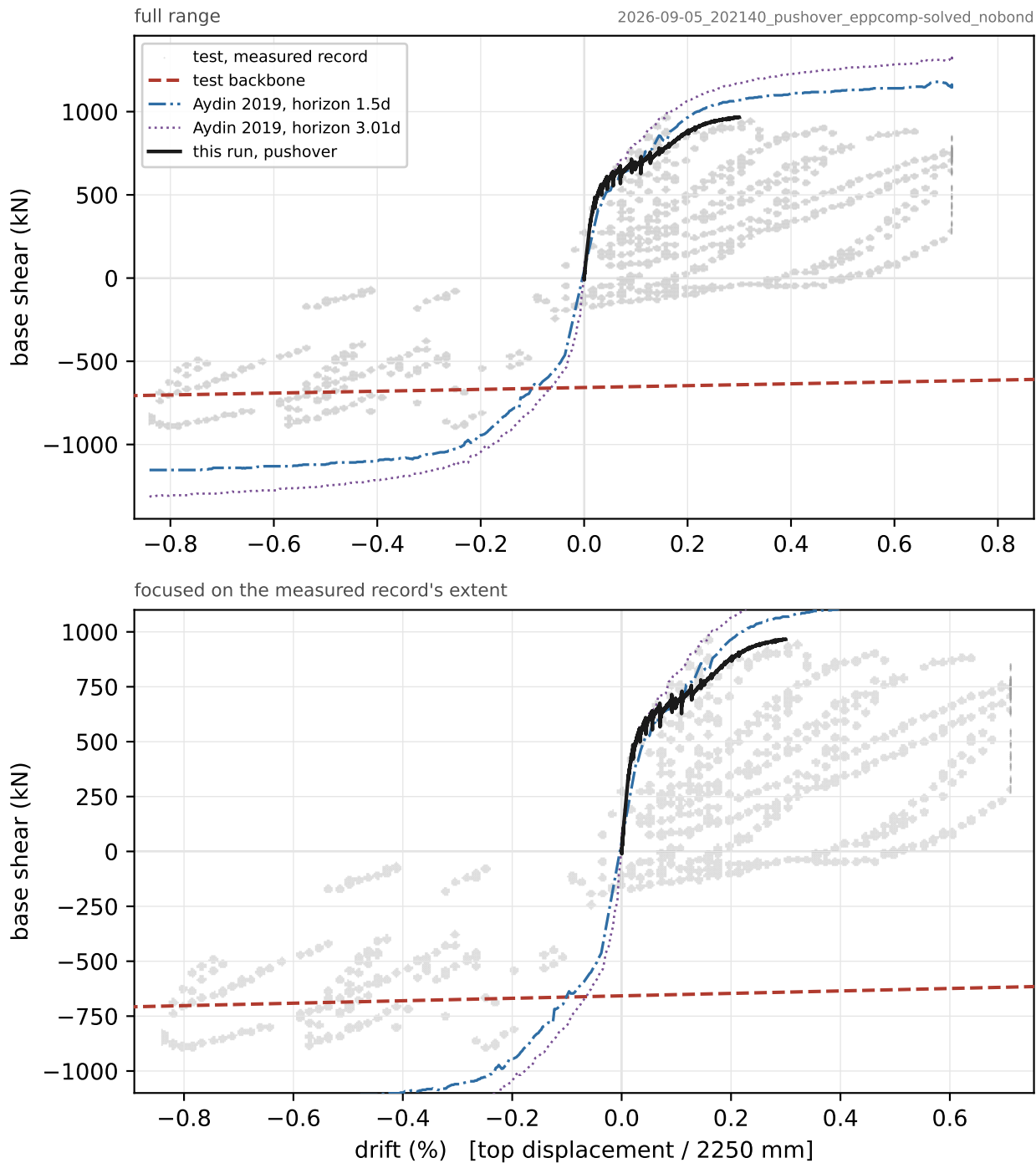


Figure 7: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3.2 Pushover — capped/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.5% drift

2026-09-05_220933_pushover_eppcomp-solved_nobond_d0.005

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	elastic-perfectly-plastic in compression, capped at f_c , no crushing
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults `bond_damage`, `steel_rupture`, `concrete_residual`, `steel_b` — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, L = 50 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\epsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis pushover

target 0.5% drift

solution scheme dynamic relaxation, quasi-static

integrator CentralDifference

time step 8 us, 728 steps per fundamental period

first period T_1 5.824 ms

drive rate 7.6 mm/s

damping 0.5 critical, mass proportional

steps 185,030

wall clock 0.34 h

converged True

Recorded response

peak base shear 995.2 kN (1 ms moving average)

raw sample maximum 1,001.2 kN

drift at peak 0.4%

drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.5% drift
inertia + damping in the reaction,	9.4 kN
ascending branch (p95)	

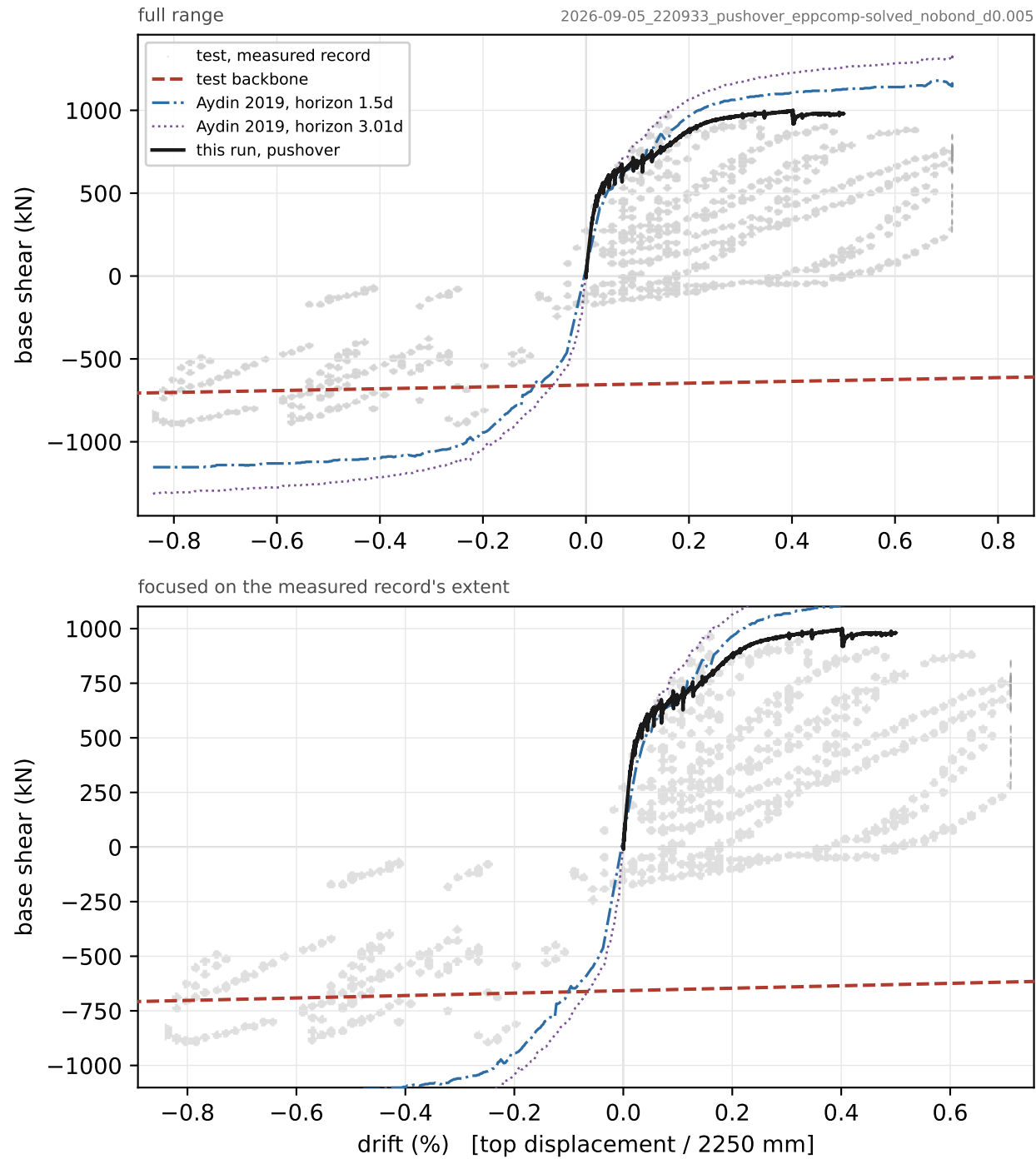


Figure 8: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

3.3 Pushover — capped/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — to 1.5% drift

2026-09-08_000752_pushover_eppcomp-solved_nobond_eps0.05_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	elastic-perfectly-plastic in compression, capped at f_c , no crushing
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	0.05 strain

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress: [-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, L = 50 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00112585, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress: [-28, -28, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis pushover

target 1.5% drift

solution scheme dynamic relaxation, quasi-static

integrator CentralDifference

time step 8 us, 728 steps per fundamental period

first period T_1 5.824 ms

drive rate 7.6 mm/s

damping 0.5 critical, mass proportional

steps 555,087

wall clock 2.31 h

converged True

Recorded response

peak base shear	1,005.6 kN (1 ms moving average)
raw sample maximum	1,010.9 kN
drift at peak	1.288%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	4.5 kN
ascending branch (p95)	

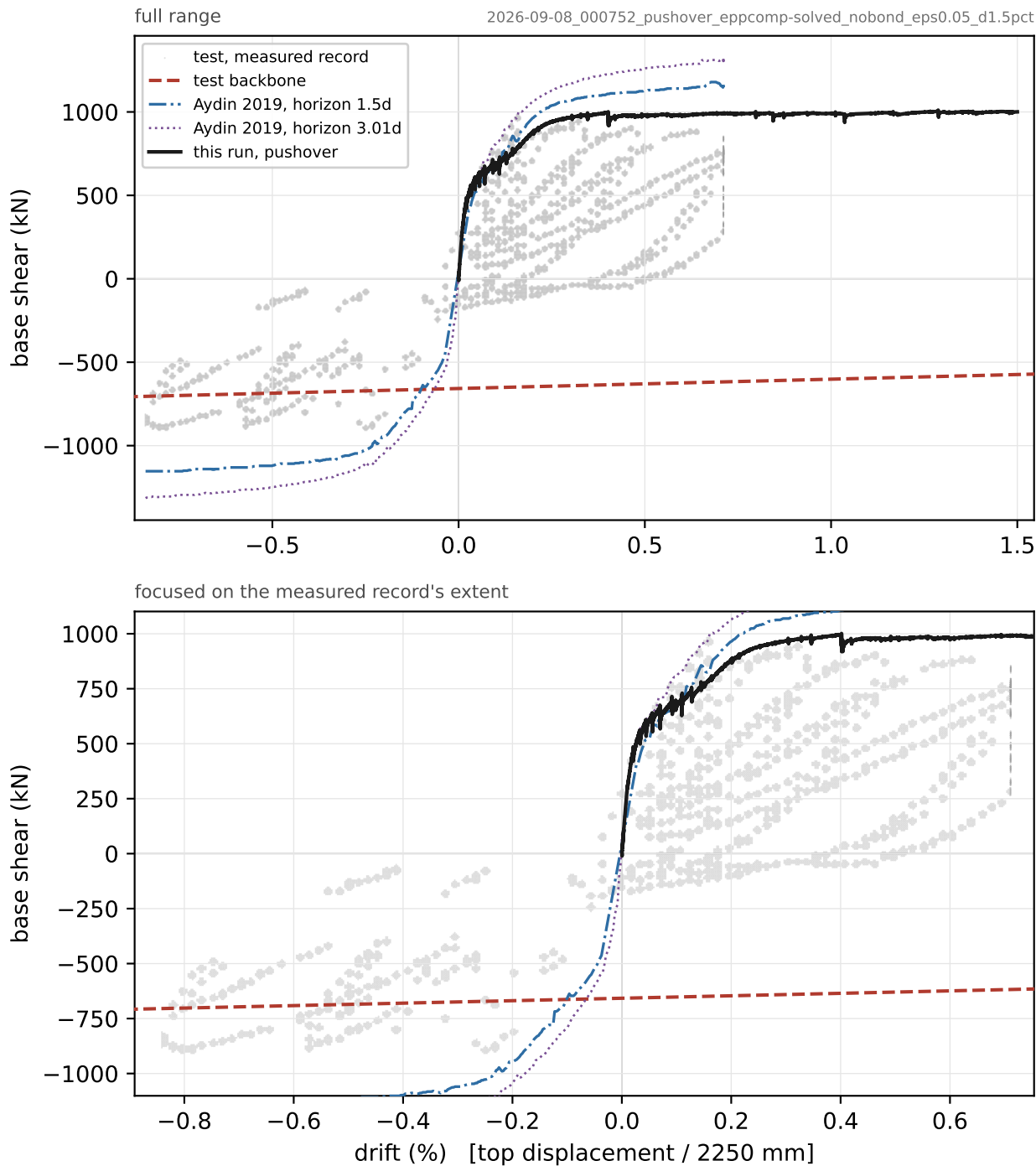


Figure 9: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

4 crushing compression — solved tension tail — bond links, $0.6f_t$ residual

4.1 Pushover — crushing/solved/bond60 — mesh 50 mm — energy balance, uniaxial field — to 1.5% drift

2026-09-07_213159_pushover_c02-solved_bond-a06_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	5,484 / 34,775
composition	5,484 nodes, 34,775 elements (21,244 bond, 10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond

tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	none — bars cannot break
bond	bond links, residual plateau $0.6 f_t$

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm
4	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,652 bond elements, L = 50 mm
5	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,592 bond elements, L = 70.71 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
bond	10,652	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
bond	10,592	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	3.58 us, 962 steps per fundamental period
first period T_1	3.444 ms

drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,240,439
wall clock	10.87 h
converged	True

Recorded response

peak base shear	1,452.4 kN (1 ms moving average)
raw sample maximum	1,488.9 kN
drift at peak	0.038%
drift capacity (20% drop)	0.051% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	12.5 kN
ascending branch (p95)	

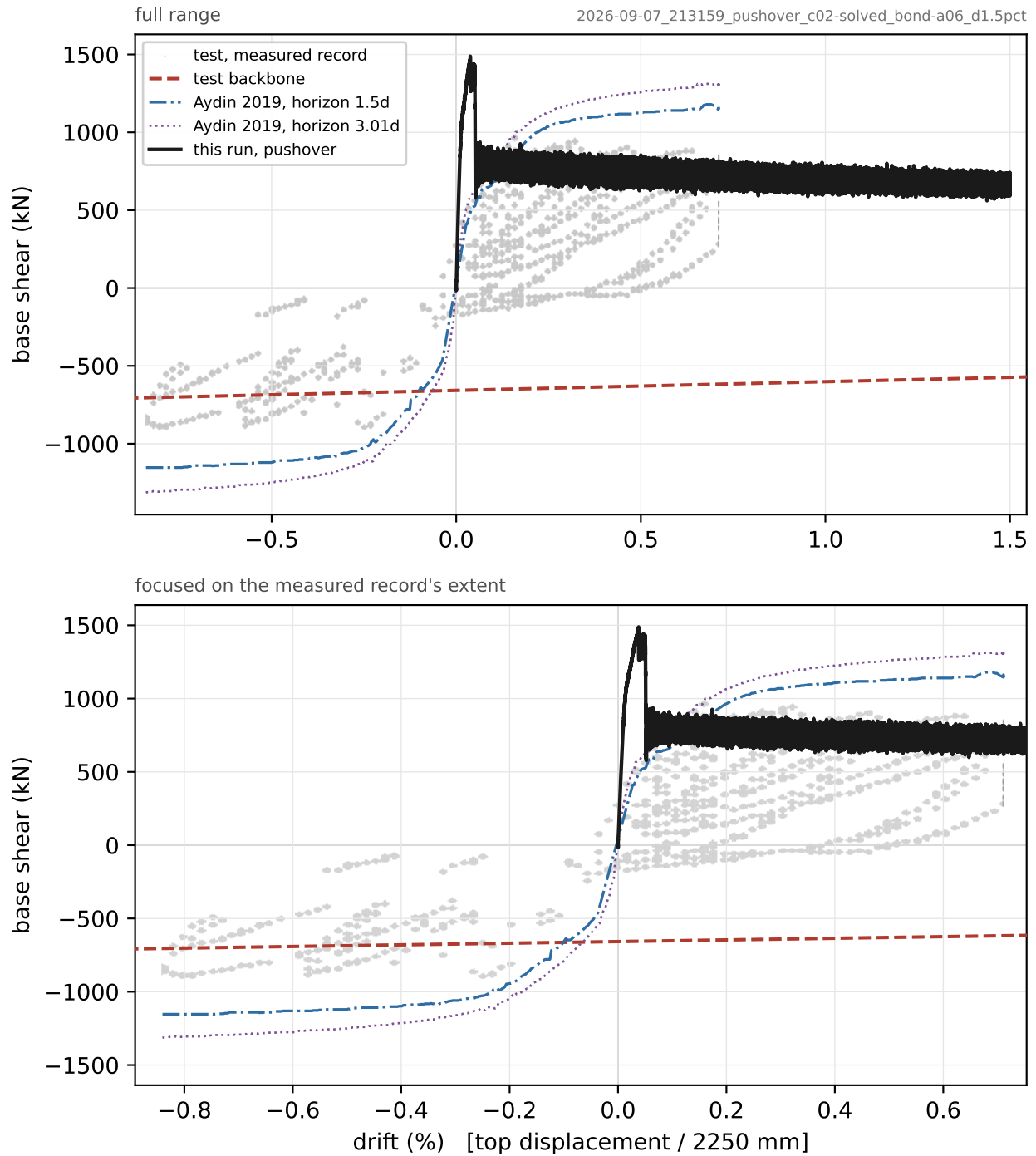


Figure 10: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

4.2 Pushover — crushing/solved/bond60 — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — to 1.5% drift

2026-09-07_213206_pushover_c02-solved_bond-a06_eps0.05_res0_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	5,484 / 34,775
composition	5,484 nodes, 34,775 elements (21,244 bond, 10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	0.05 strain
bond	bond links, residual plateau $0.6 f_t$

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm
5	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,652 bond elements, L = 50 mm
6	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,592 bond elements, L = 70.71 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
bond	10,652	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
bond	10,592	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	3.58 us, 962 steps per fundamental period
first period T_1	3.444 ms
drive rate	7.6 mm/s

damping	0.5 critical, mass proportional
steps	1,240,439
wall clock	10.83 h
converged	True

Recorded response

peak base shear	1,321.5 kN (1 ms moving average)
raw sample maximum	1,417.7 kN
drift at peak	0.045%
drift capacity (20% drop)	0.046% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	86.9 kN

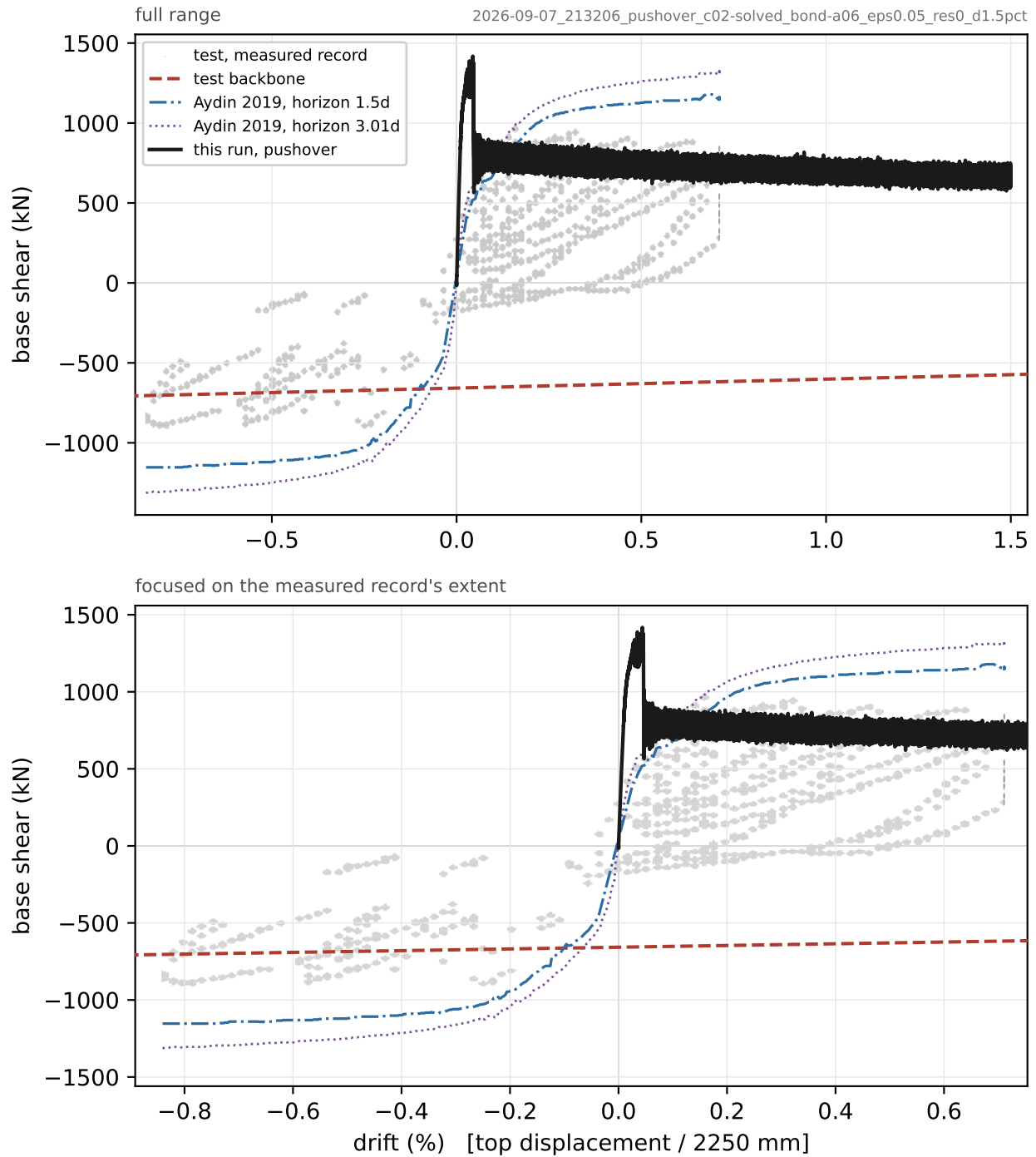


Figure 11: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

5 crushing compression — solved tension tail — bond links, $0.7f_t$ residual

5.1 Pushover — crushing/solved/bond70 — mesh 50 mm — energy balance, uniaxial field — to 1.5% drift

2026-09-08_113050_pushover_crushing-solved_bond70_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	5,484 / 34,775
composition	5,484 nodes, 34,775 elements (21,244 bond, 10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond

tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	none — bars cannot break
bond	bond links, residual plateau $0.7 f_t$

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm
4	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.295, -1.295, -1.85, 0, 1.85, 1.295, 1.295]	10,652 bond elements, L = 50 mm
5	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.295, -1.295, -1.85, 0, 1.85, 1.295, 1.295]	10,592 bond elements, L = 70.71 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
bond	10,652	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
bond	10,592	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	3.58 us, 962 steps per fundamental period
first period T_1	3.444 ms

drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,240,439
wall clock	10.23 h
converged	True

Recorded response

peak base shear	1,632.5 kN (1 ms moving average)
raw sample maximum	1,646.8 kN
drift at peak	0.043%
drift capacity (20% drop)	0.058% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	38.3 kN
ascending branch (p95)	

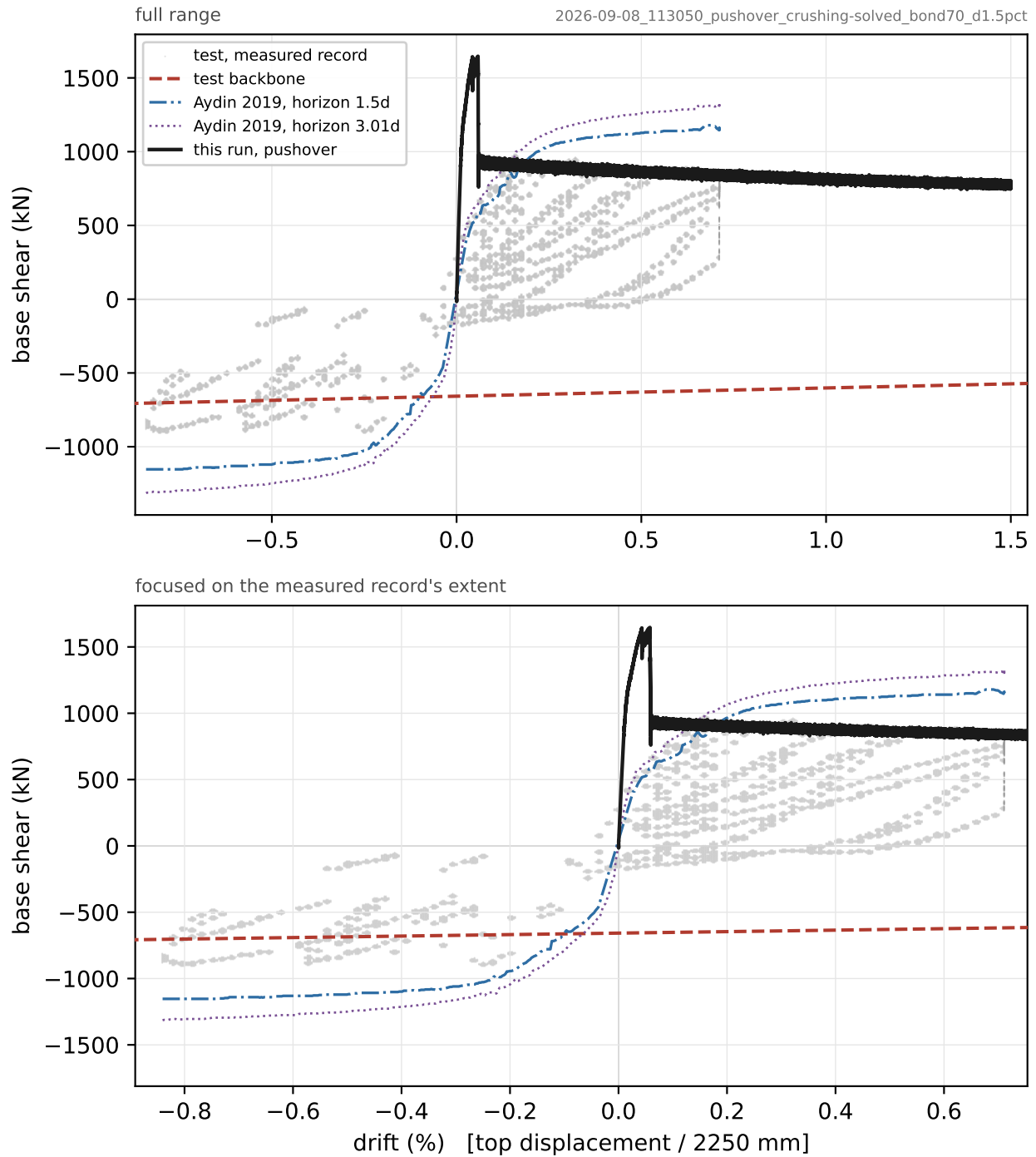


Figure 12: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

5.2 Pushover — crushing/solved/bond70 — mesh 50 mm — energy balance, uniaxial field — bond damage — to 1.5% drift

2026-09-08_113057_pushover_crushing-solved_bond70_bonddmg_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	5,484 / 34,775
composition	5,484 nodes, 34,775 elements (21,244 bond, 10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	bond links, residual plateau $0.7 f_t$

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm
4	Elastic	E:7461	referenced by tag 5
5	MinMax	otherTag:4, -min:-7.43868e-05, -max:7.43868e-05	referenced by tag 7
6	ElasticPP	E:17409, epsyP:7.43868e-05	referenced by tag 7
7	Parallel	matTag1:5, matTag2:6	10,652 bond elements, L = 50 mm
8	Elastic	E:7461	referenced by tag 9
9	MinMax	otherTag:8, -min:-7.43868e-05, -max:7.43868e-05	referenced by tag 11
10	ElasticPP	E:17409, epsyP:7.43868e-05	referenced by tag 11
11	Parallel	matTag1:9, matTag2:10	10,592 bond elements, L = 70.71 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
bond	10,652	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
bond	10,592	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static

integrator	CentralDifference
time step	3.58 us, 962 steps per fundamental period
first period T_1	3.444 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,240,439
wall clock	10.26 h
converged	True

Recorded response

peak base shear	1,775.3 kN (1 ms moving average)
raw sample maximum	1,777.7 kN
drift at peak	0.068%
drift capacity (20% drop)	0.07% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	11.1 kN

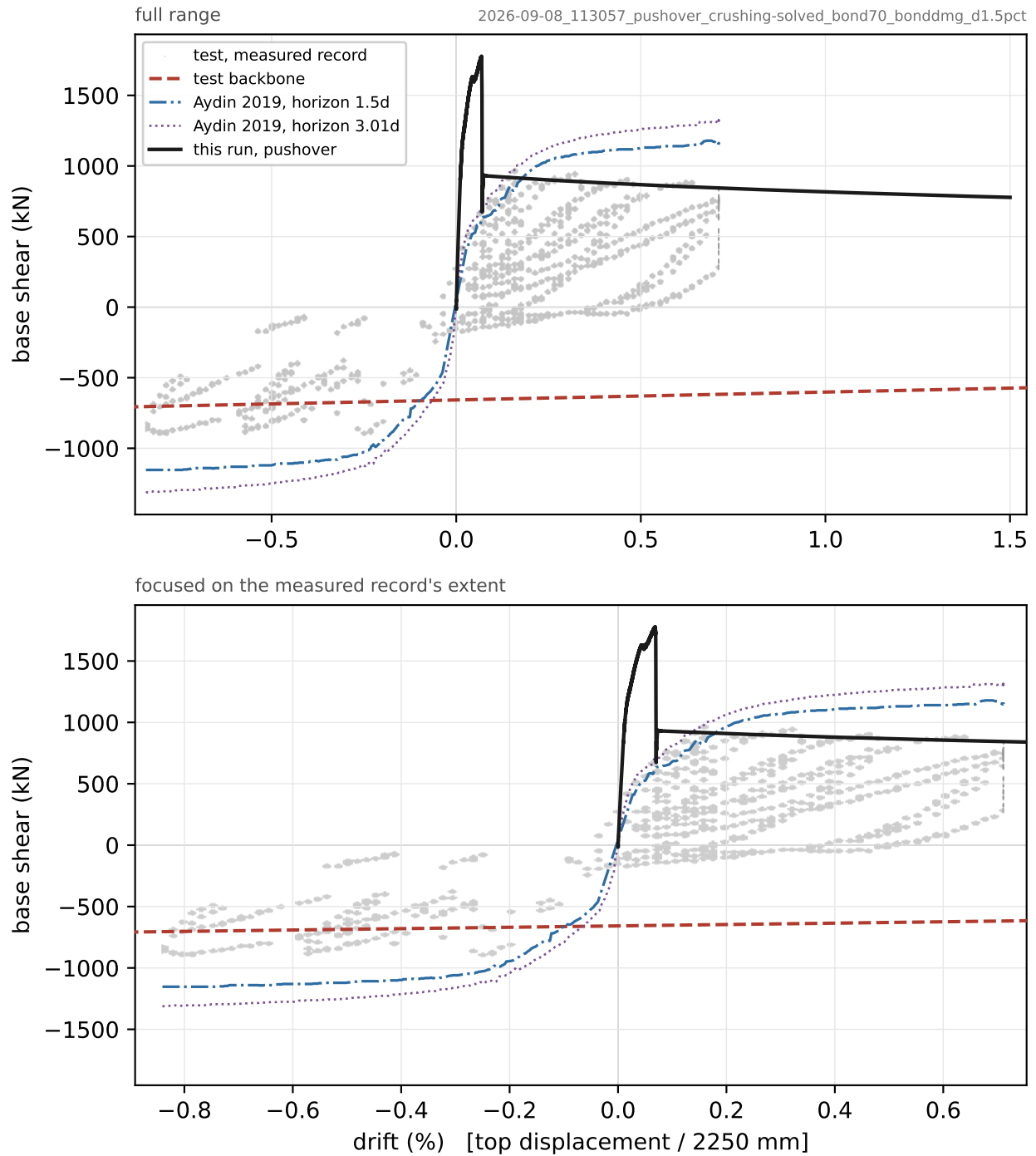


Figure 13: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6 crushing compression — solved tension tail — perfect bond

6.1 Pushover — crushing/solved/perfect — mesh 25 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — to 1.5% drift

2026-09-09_090637_pushover_crushing-solved_perfect_m25_eps0.05_res0_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	25 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	11,011 / 48,662
composition	11,011 nodes, 48,662 elements (43,410 concrete, 2,700 longitudinal, 2,552 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 44.6, diagonal 31.8
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$3,182.8 \text{ mm}^2$
$E_t A_t$ per strut	79.2e6 N
G_f multiple of the tail	1 (1.00 x G_f by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut

crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0558231, lambda:0.1, ft:1.85, Ets:570.417	21,810 concrete elements, L = 25 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0401324, lambda:0.1, ft:1.85, Ets:806.691	21,600 concrete elements, L = 35.36 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	5,252 longitudinal/stirrup elements, L = 25 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\epsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	21,810	corotTruss	25	3,182.8	24,870	7.916e7	3.166e6
concrete, diagonal	21,600	corotTruss	35.36	3,182.8	24,870	7.916e7	2.239e6
longitudinal	2,700	corotTruss	25	150.8	200,000	3.016e7	1.206e6
stirrup	2,552	corotTruss	25	150.8	200,000	3.016e7	1.206e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	3.988 us, 1,464 steps per fundamental period
first period T_1	5.838 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,113,594
wall clock	10.72 h
converged	True

Recorded response

peak base shear	1,015.1 kN (1 ms moving average)
raw sample maximum	1,016.4 kN
drift at peak	0.77%
drift capacity (20% drop)	1.108% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	6.5 kN

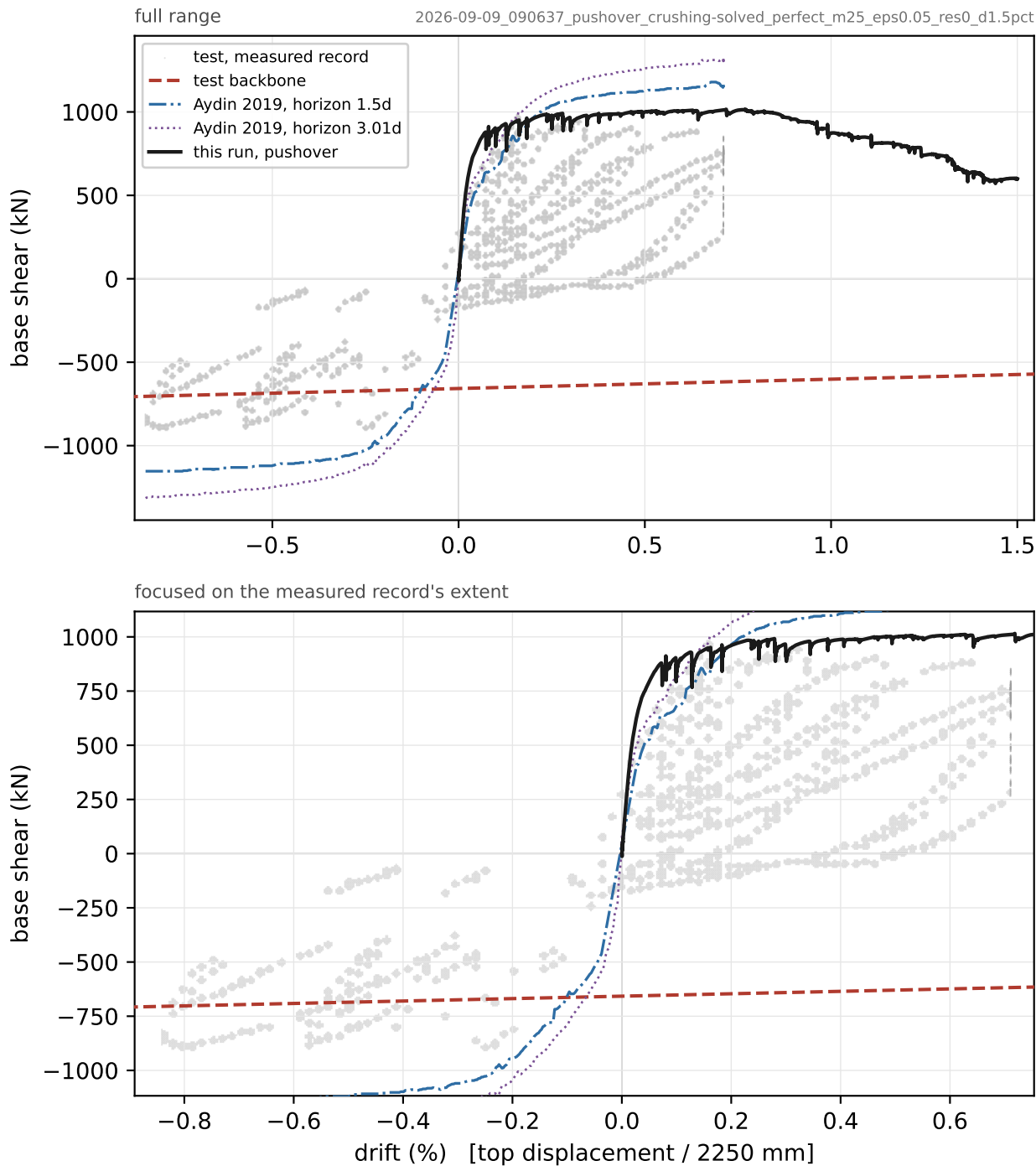


Figure 14: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.2 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — residual 0 — to 3% drift

2026-09-07_144151_pushover_c02-solved_nobond_res0_d3pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,110,173
wall clock	1.99 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	961.7 kN
drift at peak	0.646%

drift capacity (20% drop) 1.155% — basis: monotonic
traced to 3% drift
inertia + damping in the reaction, 7.7 kN
ascending branch (p95)

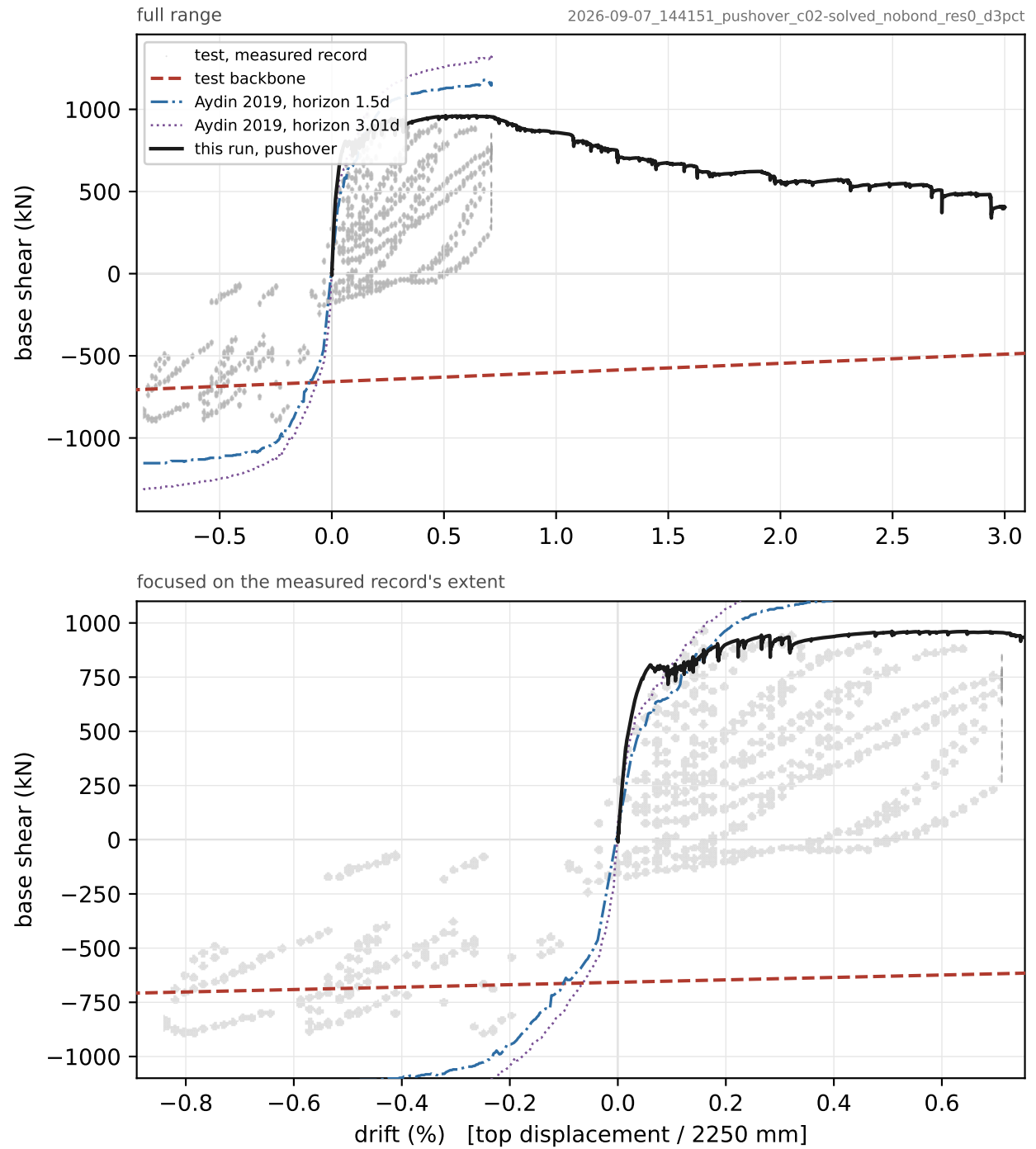


Figure 15: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.3 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.005% drift

2026-09-05_191655_pushover_c02-solved_nobond_d5e-05

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_rupture, concrete_residual, steel_b, cycles — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.005% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,851
wall clock	0 h
converged	True

Recorded response

peak base shear	150.1 kN (1 ms moving average)
raw sample maximum	155.5 kN
drift at peak	0.005%

drift capacity (20% drop) not reached — basis: monotonic
 traced to 0.005% drift
 inertia + damping in the reaction, 13.7 kN
 ascending branch (p95)

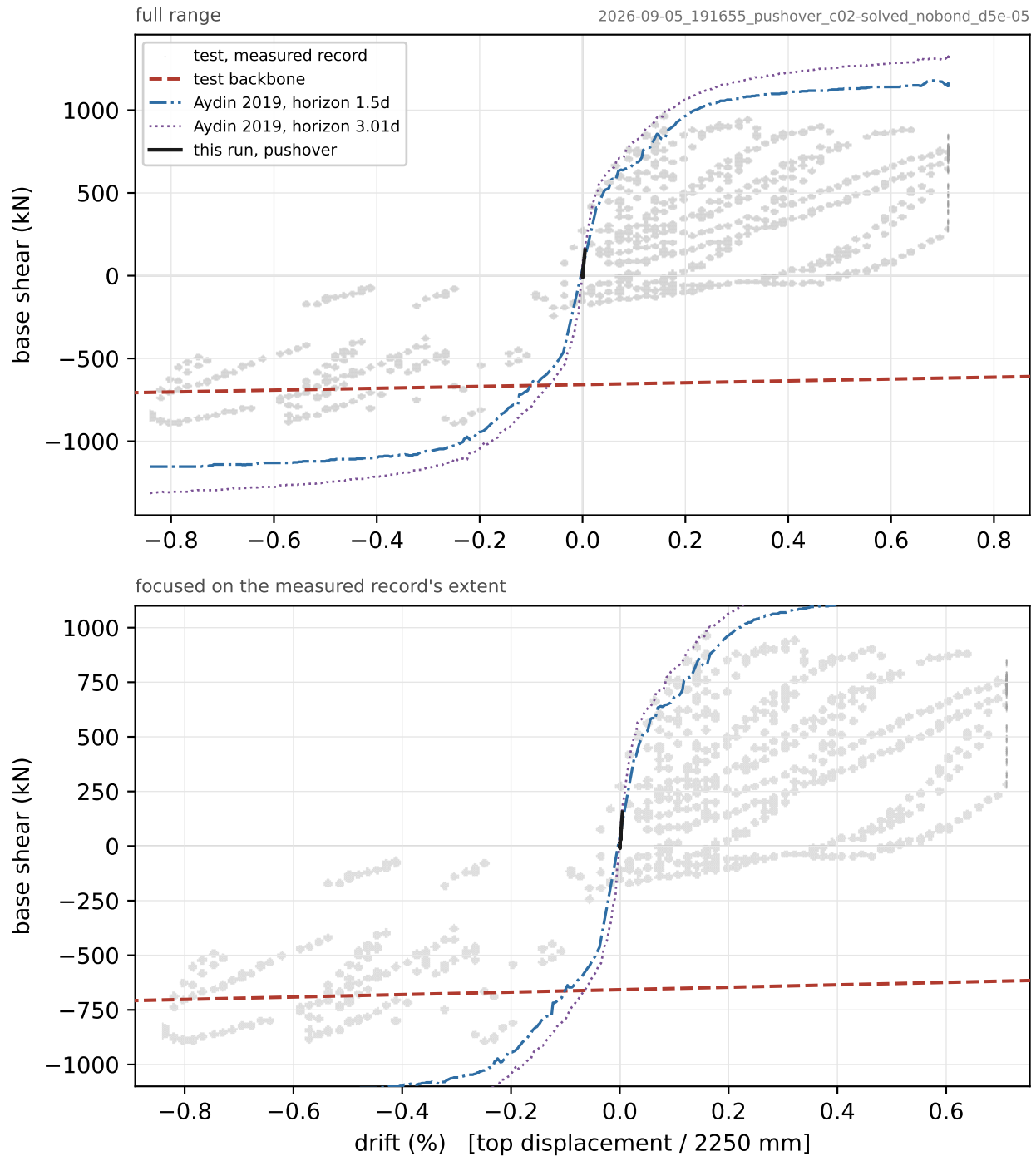


Figure 16: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.4 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.005% drift

2026-09-05_192130_pushover_c02-solved_nobond_d5e-05

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.005% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,851
wall clock	0 h
converged	True

Recorded response

peak base shear	150.1 kN (1 ms moving average)
raw sample maximum	155.5 kN
drift at peak	0.005%

drift capacity (20% drop) not reached — basis: monotonic
 traced to 0.005% drift
 inertia + damping in the reaction, 13.7 kN
 ascending branch (p95)

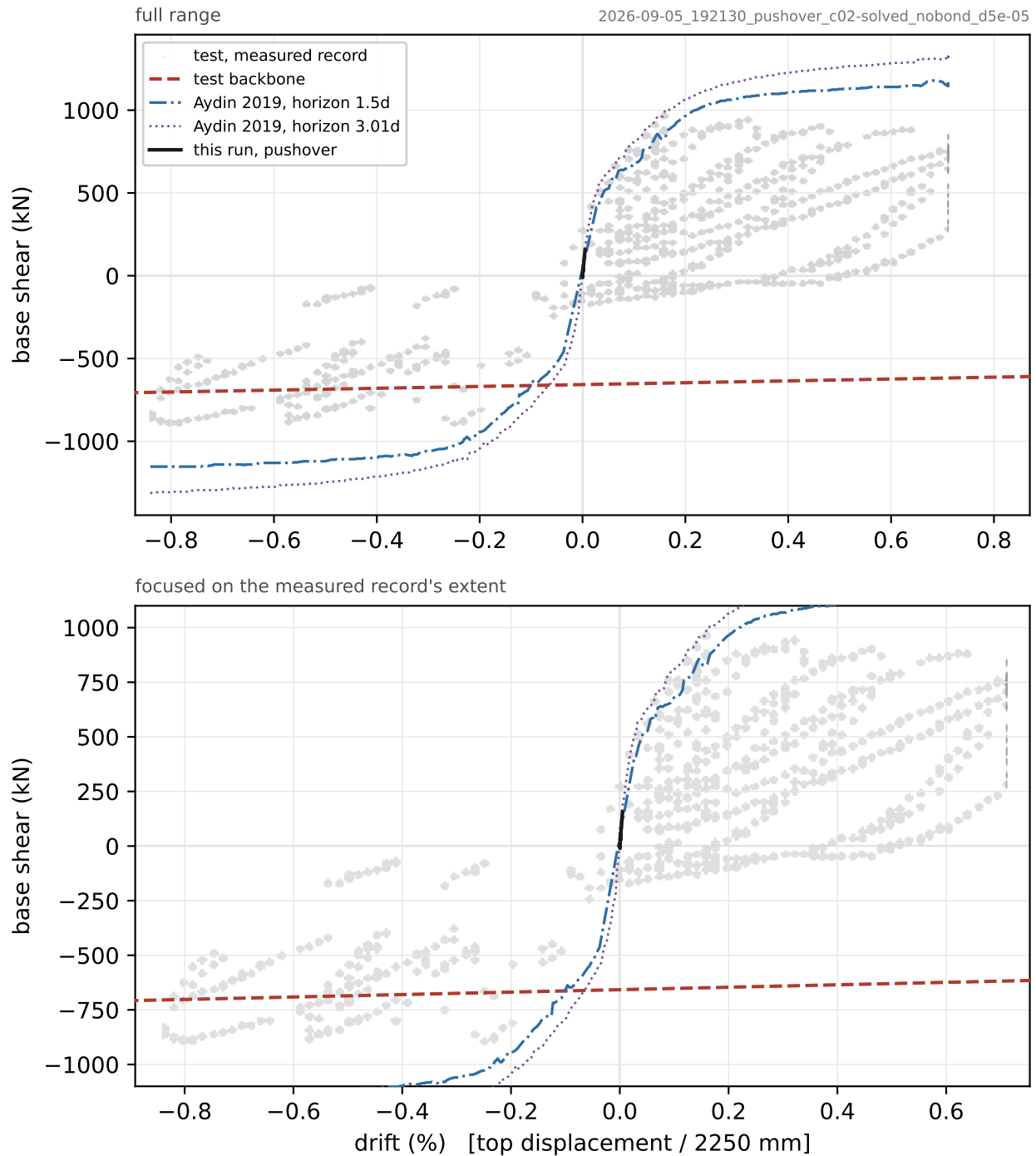


Figure 17: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.5 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_192216_pushover_c02-solved_nobond

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

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provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_rupture, concrete_residual, steel_b — this run pre-dates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μ s, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	111,018
wall clock	0.21 h
converged	True

Recorded response

peak base shear	942.4 kN (1 ms moving average)
raw sample maximum	943.8 kN
drift at peak	0.281%

drift capacity (20% drop) not reached — basis: monotonic
 traced to 0.3% drift
 inertia + damping in the reaction, 10.4 kN
 ascending branch (p95)

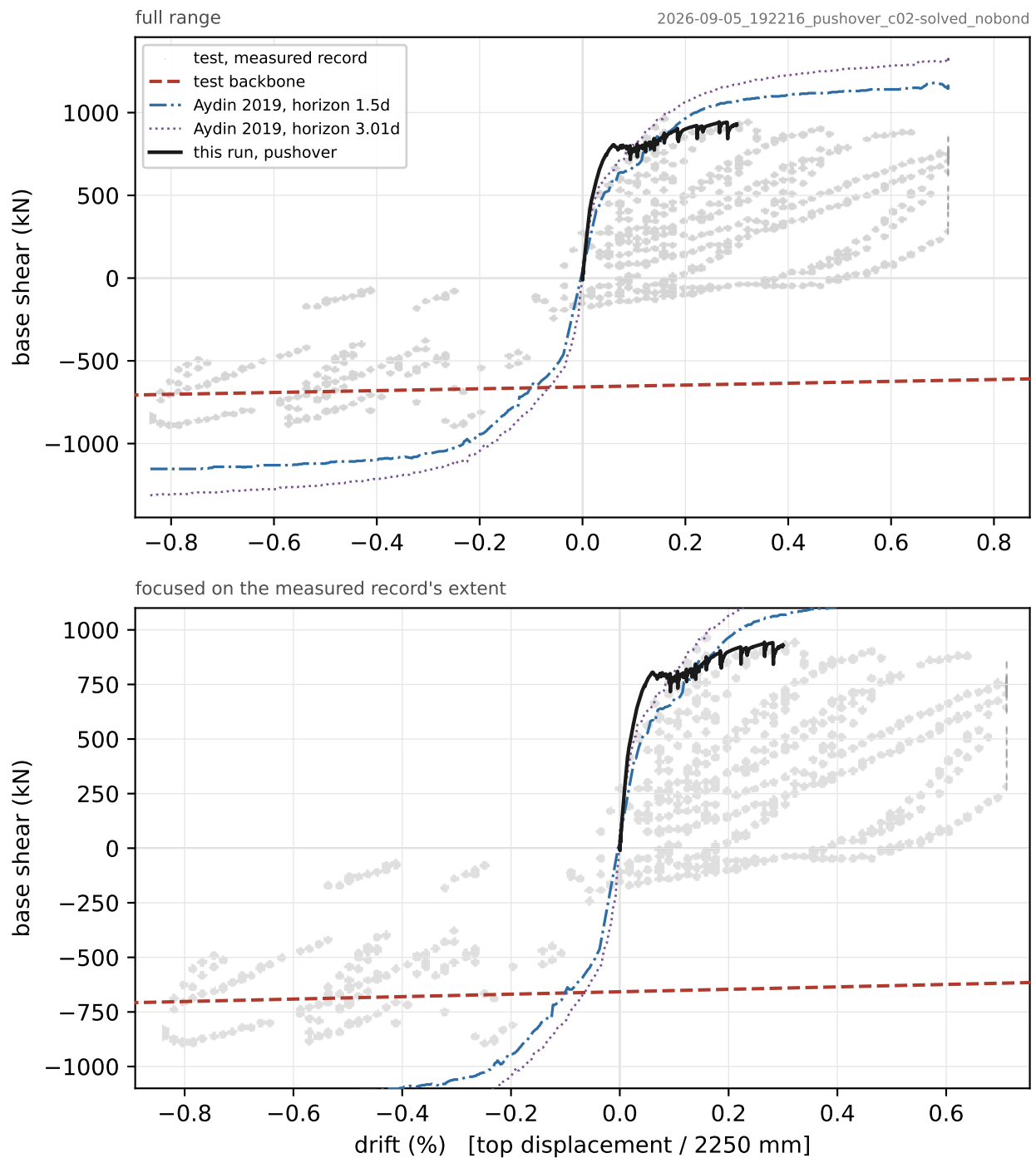


Figure 18: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.6 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.025$ — residual 0 — $b = 0$ — to 1.5% drift

2026-09-12_104501_pushover_crushing-solved_perfect_eps0.025_res0_sb0_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0$

bar rupture	0.025 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	0.99 h
converged	True

Recorded response

peak base shear	926.4 kN (1 ms moving average)
raw sample maximum	954.7 kN
drift at peak	0.269%
drift capacity (20% drop)	0.308% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	18.7 kN

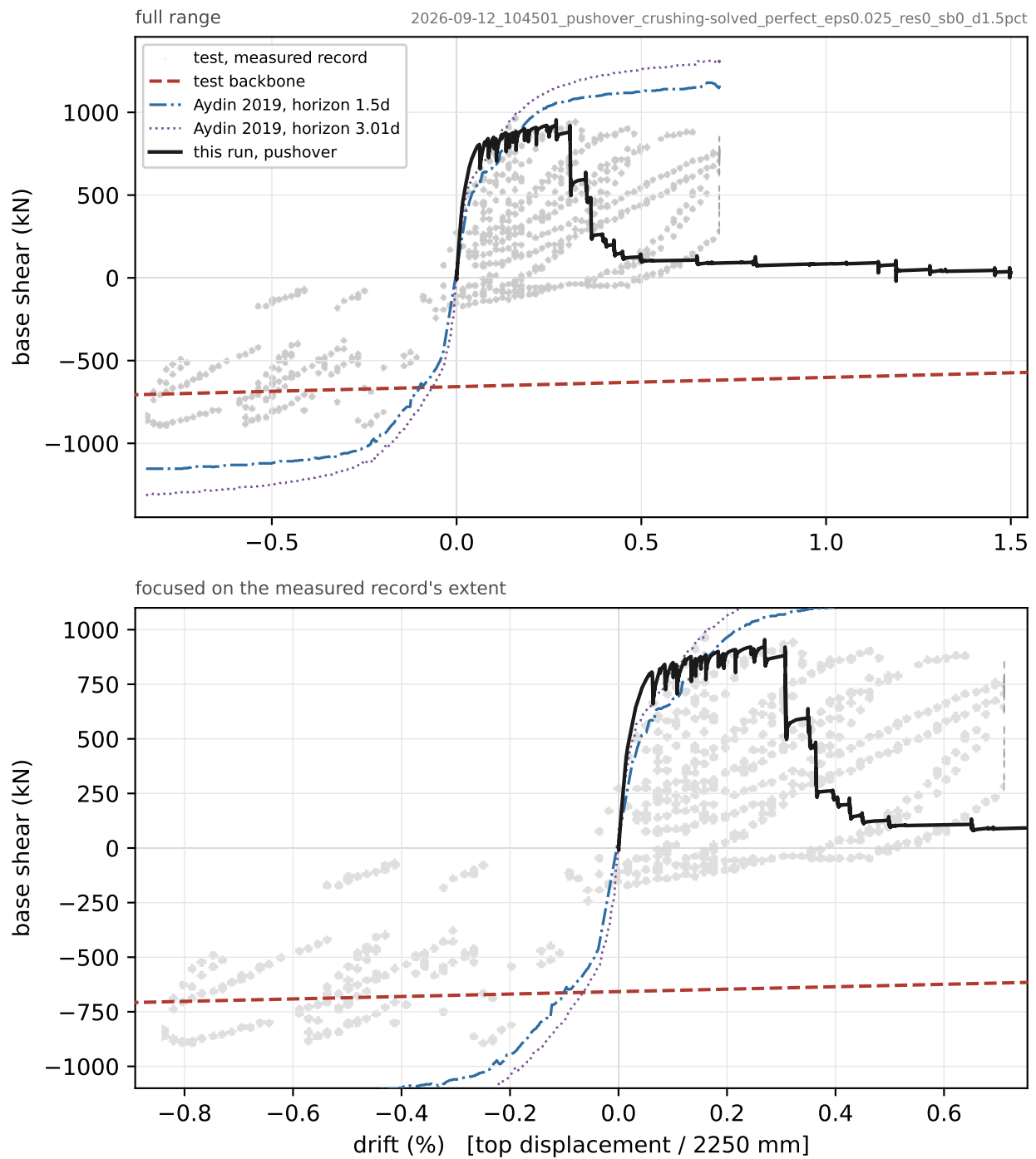


Figure 19: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.7 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.025$ — residual 0 — $b = 0.0001$ — to 1.5% drift

2026-09-11_164608_pushover_crushing-solved_perfect_eps0.025_res0_sb0.0001_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.0001$

bar rupture	0.025 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
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16	MinMax	otherTag:15, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
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20	MinMax	otherTag:19, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
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23	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
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28	MinMax	otherTag:27, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
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30	MinMax	otherTag:29, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
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34	MinMax	otherTag:33, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.17 h
converged	True

Recorded response

peak base shear	926.6 kN (1 ms moving average)
raw sample maximum	955 kN
drift at peak	0.269%
drift capacity (20% drop)	0.311% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	18.6 kN

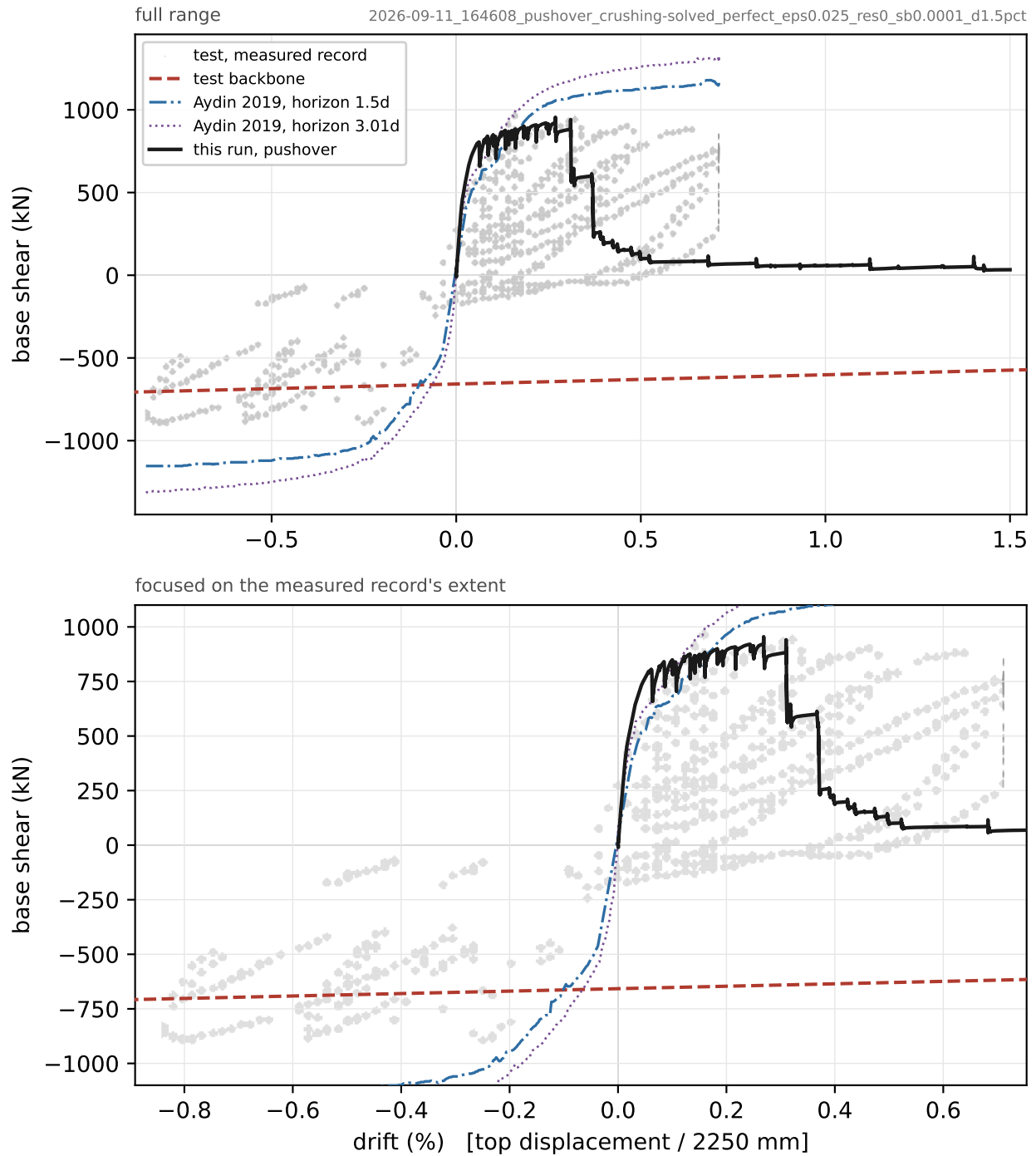


Figure 20: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.8 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.025$ — residual 0 — $b = 0.001$ — to 1.5% drift

2026-09-11_164608_pushover_crushing-solved_perfect_eps0.025_res0_sb0.001_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x G_f by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.001$

bar rupture	0.025 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
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tag	type	arguments	applied to
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34	MinMax	otherTag:33, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

[illegible]

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.11 h
converged	True

Recorded response

peak base shear	929.8 kN (1 ms moving average)
raw sample maximum	959.1 kN
drift at peak	0.283%
drift capacity (20% drop)	0.365% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	13.6 kN

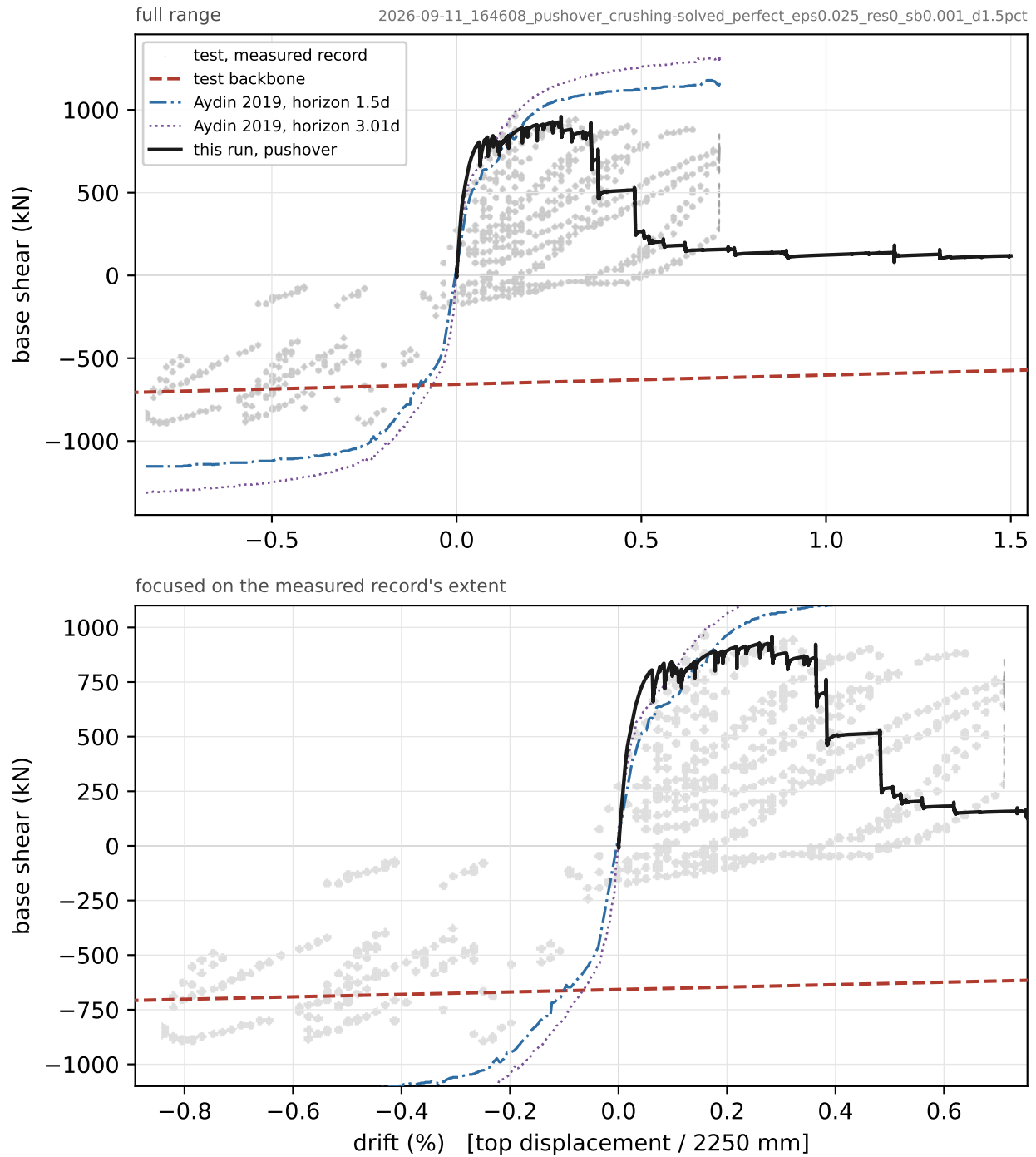


Figure 21: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.9 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.025$ — residual 0 — $b = 0.0036$ — to 1.5% drift

2026-09-11_164608_pushover_crushing-solved_perfect_eps0.025_res0_sb0.0036_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.0036$

bar rupture	0.025 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.025, -max:0.025	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0.0036, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.025, -max:0.025	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

[illegible]

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.18 h
converged	True

Recorded response

peak base shear	937.2 kN (1 ms moving average)
raw sample maximum	950.6 kN
drift at peak	0.327%
drift capacity (20% drop)	0.48% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	10 kN

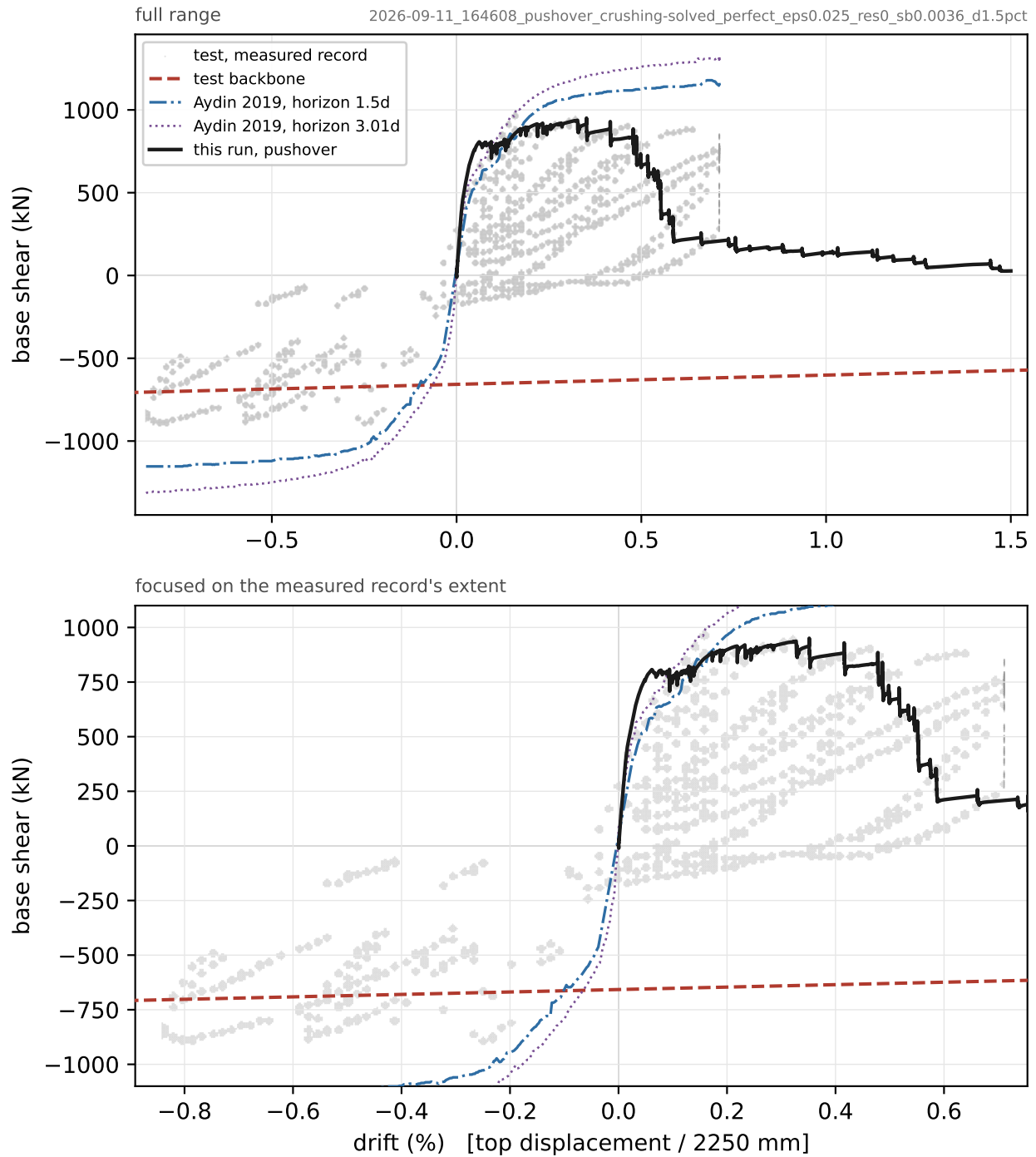


Figure 22: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.10 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.025$ — residual 0 — to 3% drift

2026-09-07_184324_pushover_c02-solved_nobond_eps0.025_res0_d3pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	0.025 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.025, -max:0.025	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,110,173
wall clock	1.99 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	996.6 kN
drift at peak	0.469%
drift capacity (20% drop)	0.592% — basis: monotonic
traced to	3% drift
inertia + damping in the reaction, ascending branch (p95)	8.4 kN

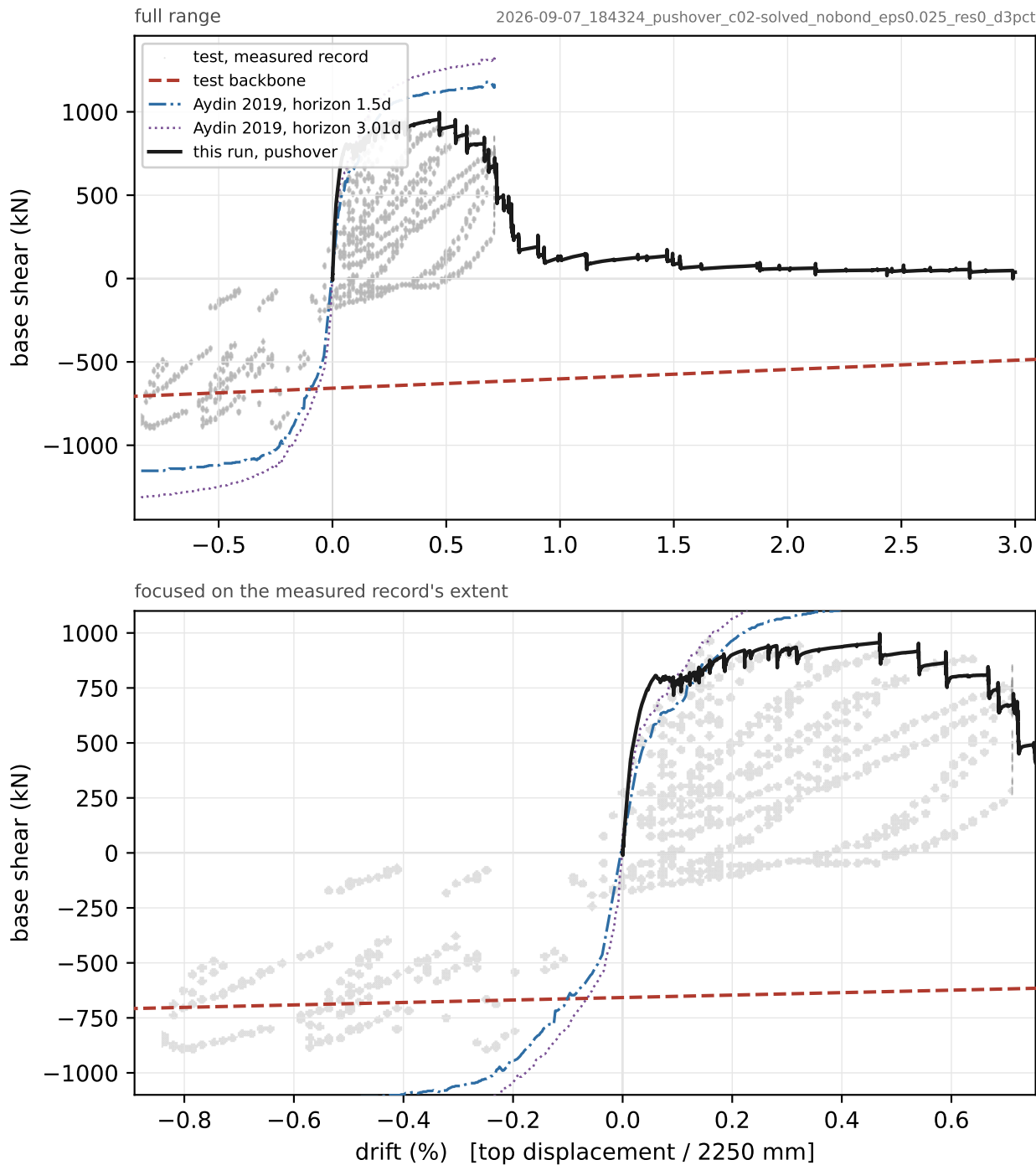


Figure 23: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.11 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $b = 0$ — to 1.5% drift

2026-09-11_110810_pushover_crushing-solved_perfect_eps0.05_res0_sb0_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.2 h
converged	True

Recorded response

peak base shear	929 kN (1 ms moving average)
raw sample maximum	954.2 kN
drift at peak	0.327%
drift capacity (20% drop)	0.476%
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	15.8 kN

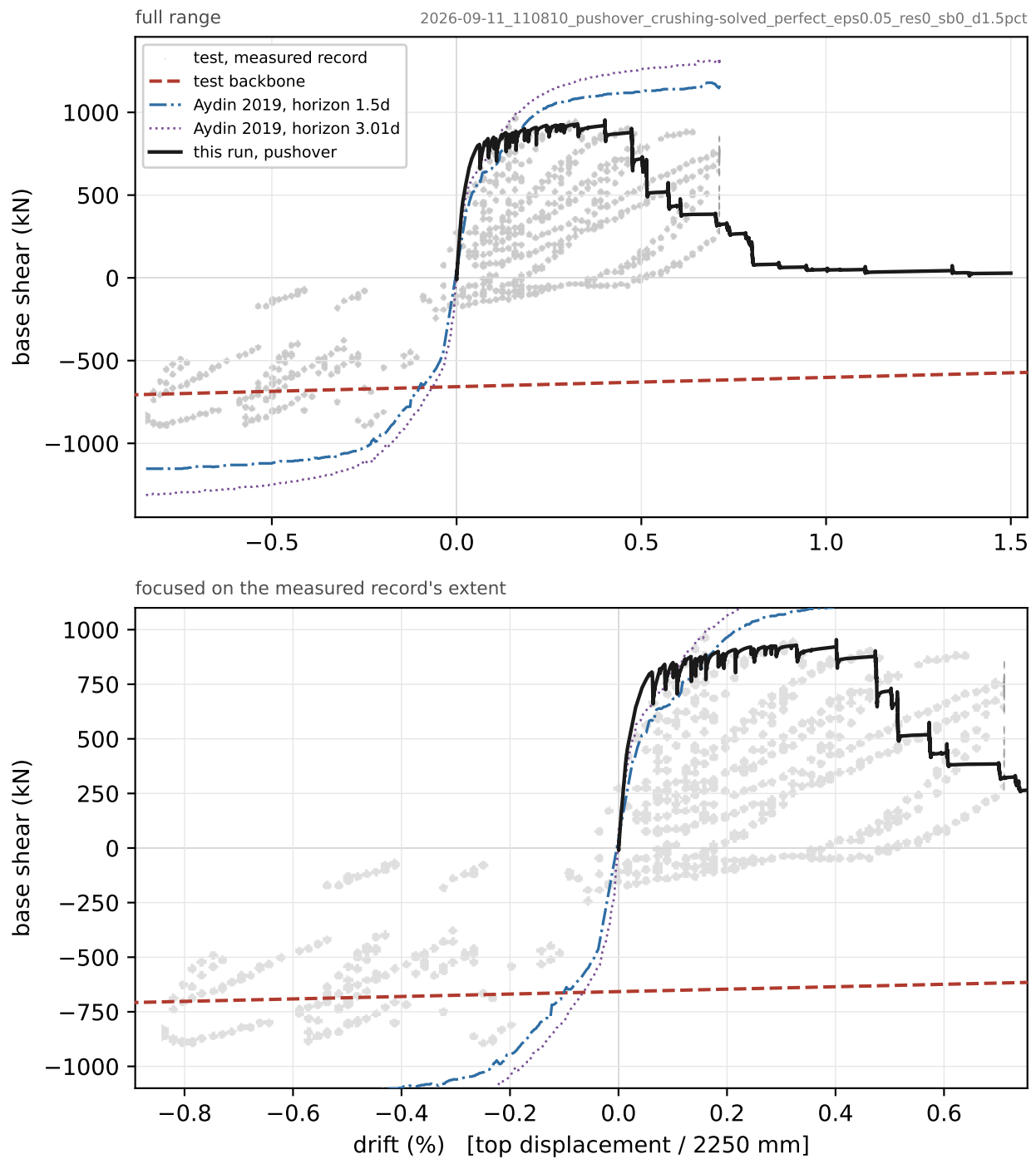


Figure 24: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.12 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $b = 0.0001$ — to 1.5% drift

2026-09-11_110810_pushover_crushing-solved_perfect_eps0.05_res0_sb0.0001_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.0001$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0.0001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.24 h
converged	True

Recorded response

peak base shear	929.9 kN (1 ms moving average)
raw sample maximum	953.5 kN
drift at peak	0.328%
drift capacity (20% drop)	0.476%
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	16.2 kN

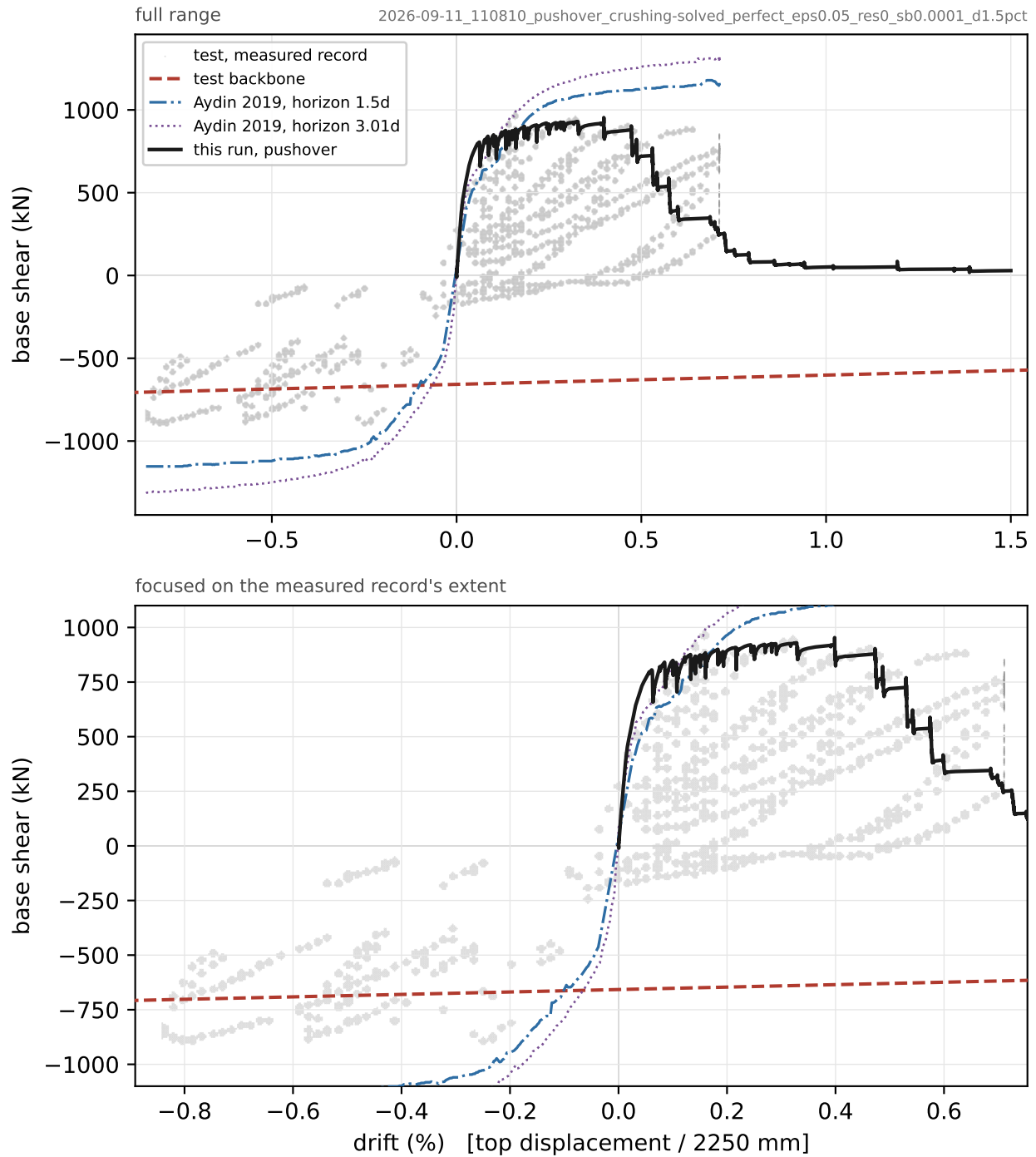


Figure 25: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.13 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $b = 0.001$ — to 1.5% drift

2026-09-11_110810_pushover_crushing-solved_perfect_eps0.05_res0_sb0.001_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x G_f by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.001$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
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tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
5	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 6
6	MinMax	otherTag:5, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
7	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 8
8	MinMax	otherTag:7, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
9	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 10
10	MinMax	otherTag:9, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
11	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 12
12	MinMax	otherTag:11, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
13	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 14
14	MinMax	otherTag:13, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
15	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 16
16	MinMax	otherTag:15, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
17	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 18
18	MinMax	otherTag:17, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
19	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 20
20	MinMax	otherTag:19, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
21	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 22
22	MinMax	otherTag:21, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
23	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 24
24	MinMax	otherTag:23, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm

tag	type	arguments	applied to
25	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 26
26	MinMax	otherTag:25, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
27	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 28
28	MinMax	otherTag:27, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
29	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 30
30	MinMax	otherTag:29, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
31	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 32
32	MinMax	otherTag:31, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
33	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 34
34	MinMax	otherTag:33, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
35	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 36
36	MinMax	otherTag:35, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
37	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 38
38	MinMax	otherTag:37, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
39	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 40
40	MinMax	otherTag:39, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
41	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 42
42	MinMax	otherTag:41, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
43	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 44
44	MinMax	otherTag:43, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
45	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 46
46	MinMax	otherTag:45, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
47	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 48
48	MinMax	otherTag:47, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
49	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 50
50	MinMax	otherTag:49, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
51	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 52
52	MinMax	otherTag:51, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
53	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 54

tag	type	arguments	applied to
54	MinMax	otherTag:53, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
55	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 56
56	MinMax	otherTag:55, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
57	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 58
58	MinMax	otherTag:57, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
59	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 60
60	MinMax	otherTag:59, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
61	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 62
62	MinMax	otherTag:61, -min:-0.05, -max:0.05	45 longitudinal elements, L = 50 mm
63	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 64
64	MinMax	otherTag:63, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
65	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 66
66	MinMax	otherTag:65, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
67	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 68
68	MinMax	otherTag:67, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
69	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 70
70	MinMax	otherTag:69, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
71	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 72
72	MinMax	otherTag:71, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
73	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 74
74	MinMax	otherTag:73, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
75	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 76
76	MinMax	otherTag:75, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
77	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 78
78	MinMax	otherTag:77, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
79	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 80
80	MinMax	otherTag:79, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
81	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 82
82	MinMax	otherTag:81, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

tag	type	arguments	applied to
83	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 84
84	MinMax	otherTag:83, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
85	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 86
86	MinMax	otherTag:85, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
87	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 88
88	MinMax	otherTag:87, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
89	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 90
90	MinMax	otherTag:89, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
91	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 92
92	MinMax	otherTag:91, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
93	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 94
94	MinMax	otherTag:93, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
95	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 96
96	MinMax	otherTag:95, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
97	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 98
98	MinMax	otherTag:97, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
99	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 100
100	MinMax	otherTag:99, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
101	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 102
102	MinMax	otherTag:101, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
103	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 104
104	MinMax	otherTag:103, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm
105	Steel02	Fy:360, E0:200000, b:0.001, R0:18, cR1:0.925, cR2:0.15	referenced by tag 106
106	MinMax	otherTag:105, -min:-0.05, -max:0.05	58 stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	0 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.24 h
converged	True

Recorded response

peak base shear	936.8 kN (1 ms moving average)
raw sample maximum	967.1 kN
drift at peak	0.442%
drift capacity (20% drop)	0.575%
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	12.1 kN

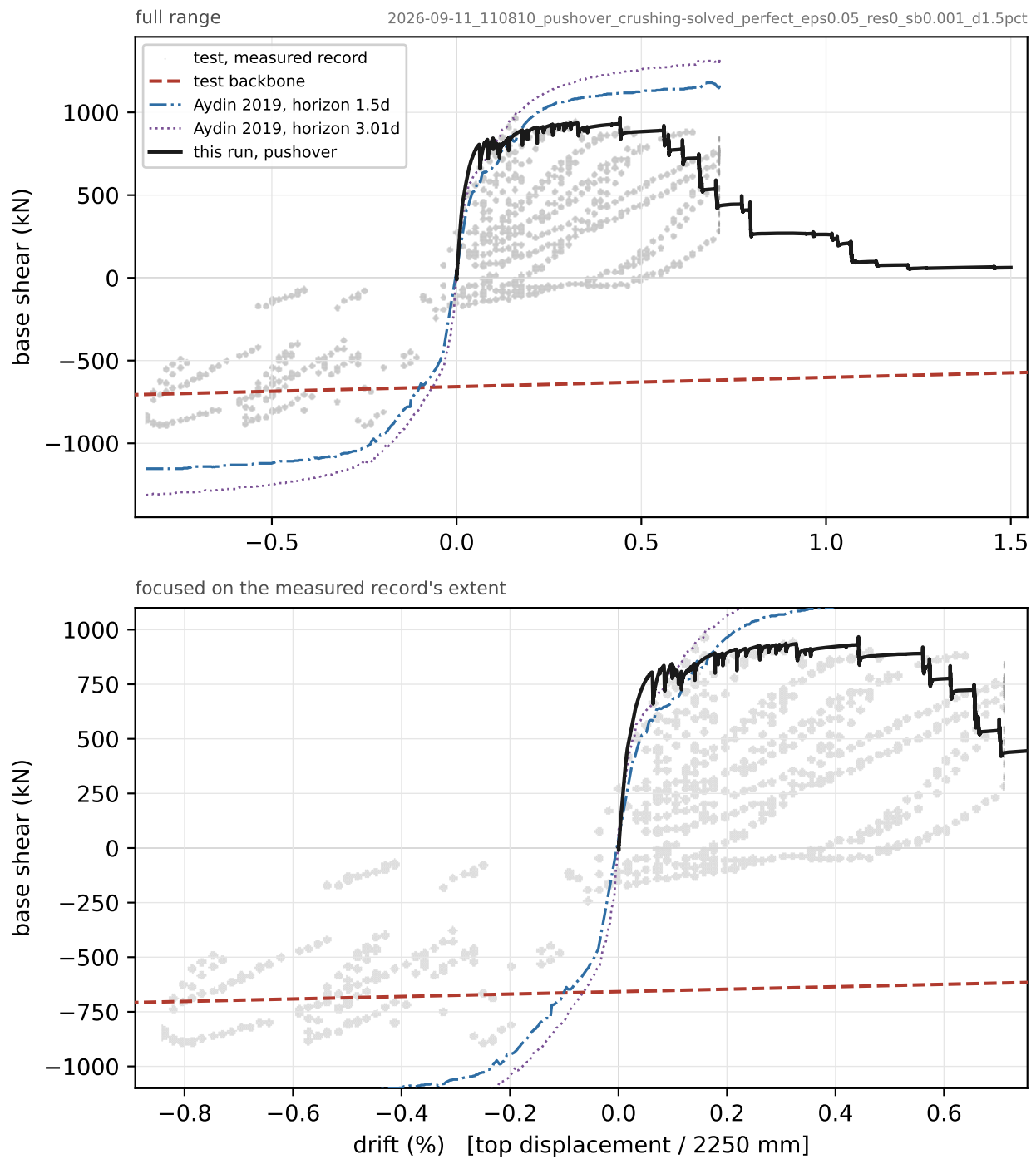


Figure 26: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.14 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $\zeta = 0.05$ — to 1.5% drift

2026-09-08_214959_pushover_crushing-solved_perfect_eps0.05_res0_d1.5pct_z0.05

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.05 critical, mass proportional
steps	555,087
wall clock	2.22 h
converged	True

Recorded response

peak base shear	964.5 kN (1 ms moving average)
raw sample maximum	967.5 kN
drift at peak	0.605%
drift capacity (20% drop)	1.113% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction, ascending branch (p95)	25.9 kN

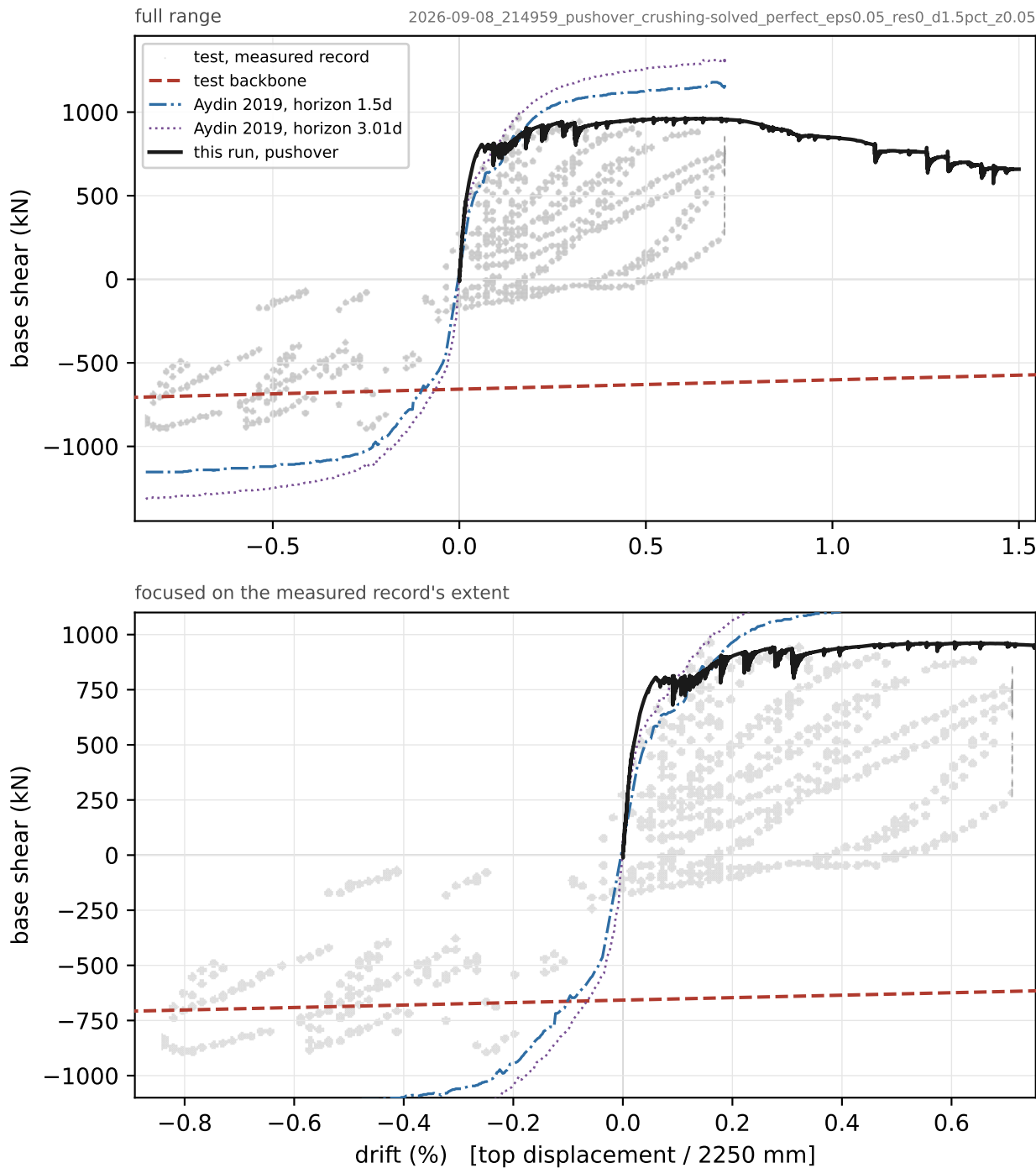


Figure 27: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.15 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — $\zeta = 0.2$ — to 1.5% drift

2026-09-08_215004_pushover_crushing-solved_perfect_eps0.05_res0_d1.5pct_z0.2

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.2 critical, mass proportional
steps	555,087
wall clock	2.26 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	961.9 kN
drift at peak	0.607%
drift capacity (20% drop)	1.152% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	15.3 kN
ascending branch (p95)	

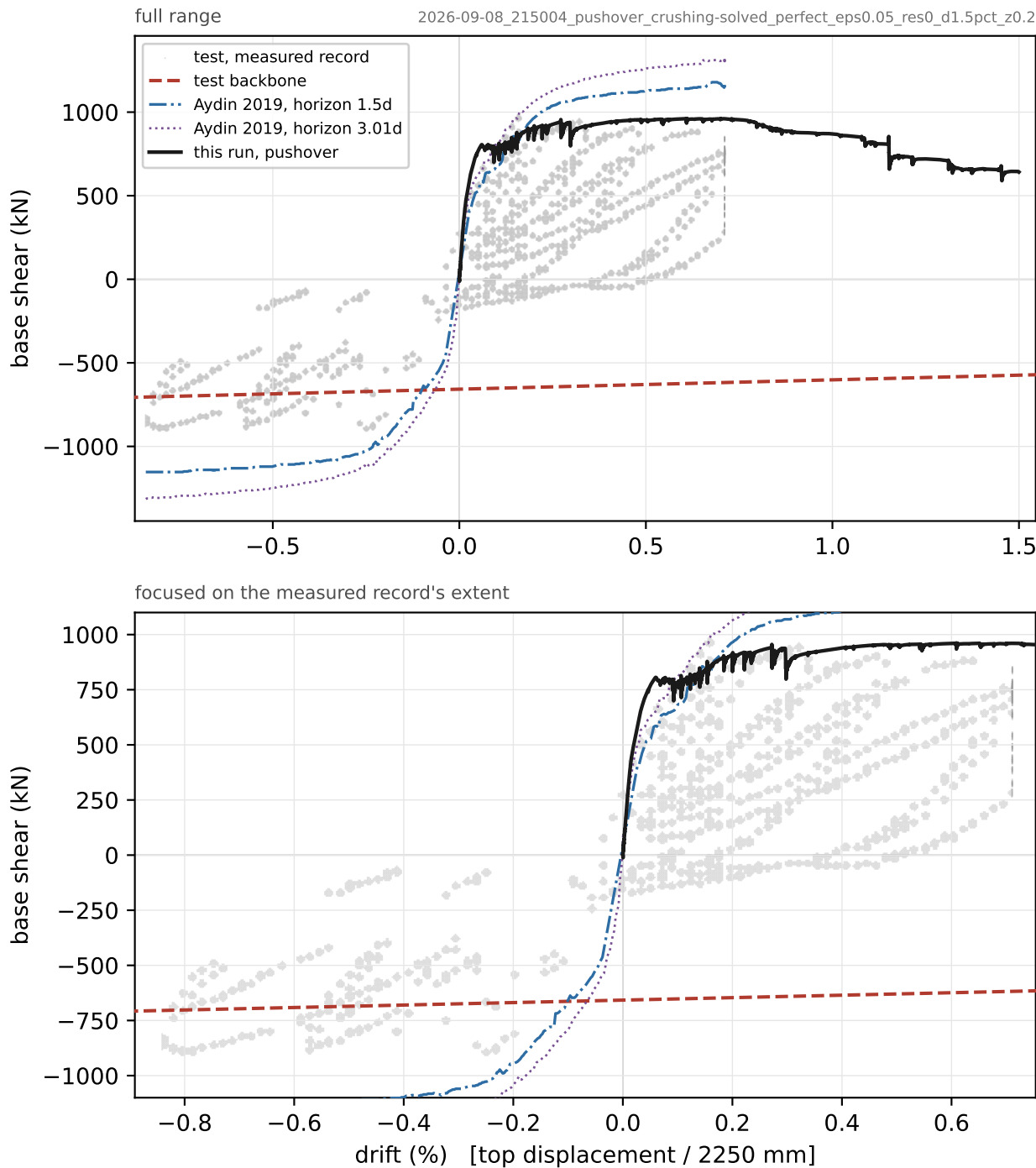


Figure 28: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.16 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — to 1.5% drift

2026-09-08_215009_pushover_crushing-solved_perfect_eps0.05_res0_d1.5pct

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>steel_b</code> — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	1.5% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	555,087
wall clock	2.28 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	961.7 kN
drift at peak	0.646%
drift capacity (20% drop)	1.033% — basis: monotonic
traced to	1.5% drift
inertia + damping in the reaction,	7.7 kN
ascending branch (p95)	

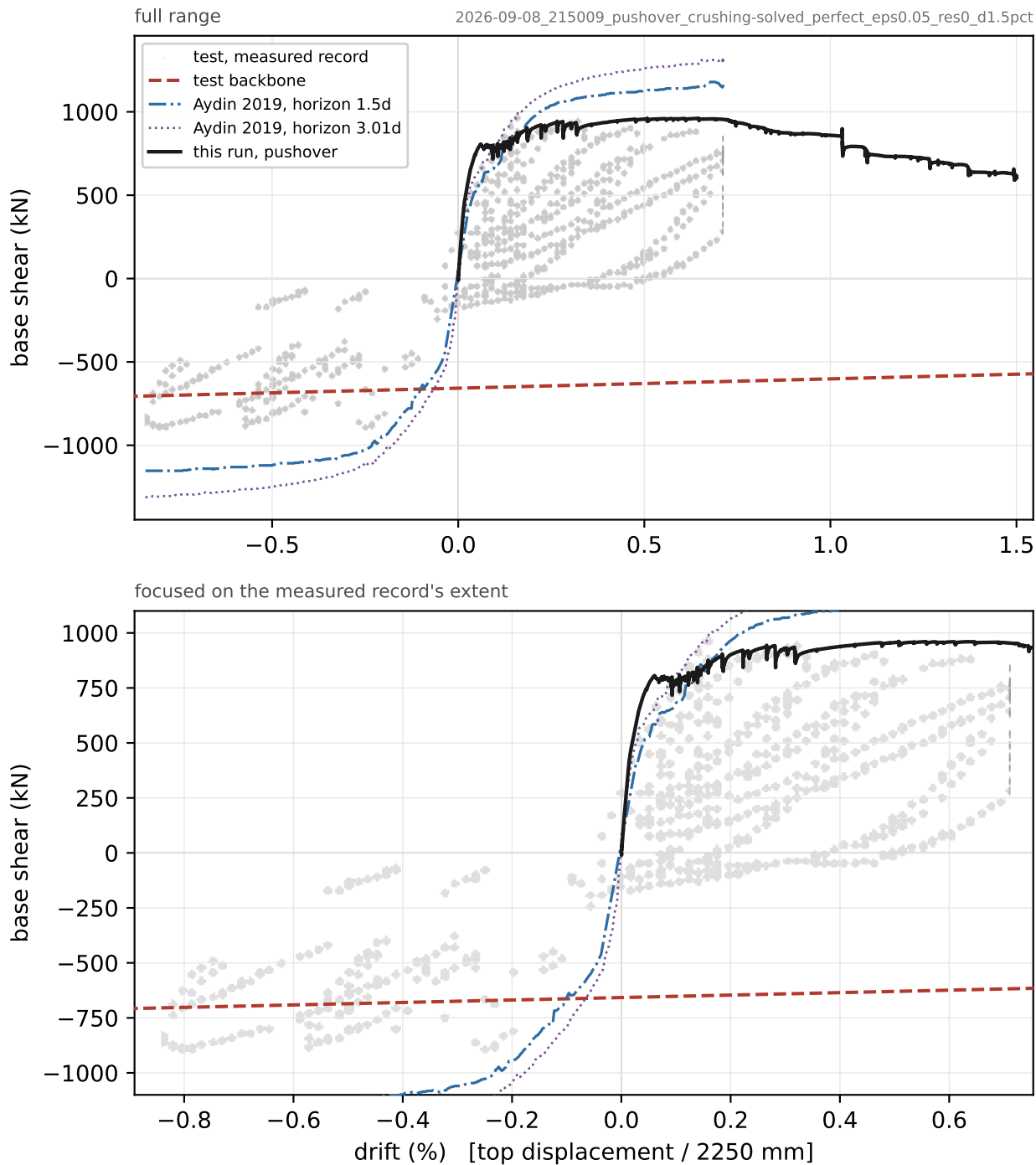


Figure 29: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.17 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — residual 0 — to 2% drift

2026-09-07_082816_pushover_c02-solved_nobond_eps0.05_res0_d2pct

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	2% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	740,116
wall clock	2.35 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	961.7 kN
drift at peak	0.646%
drift capacity (20% drop)	1.033% — basis: monotonic
traced to	2% drift
inertia + damping in the reaction,	7.7 kN
ascending branch (p95)	

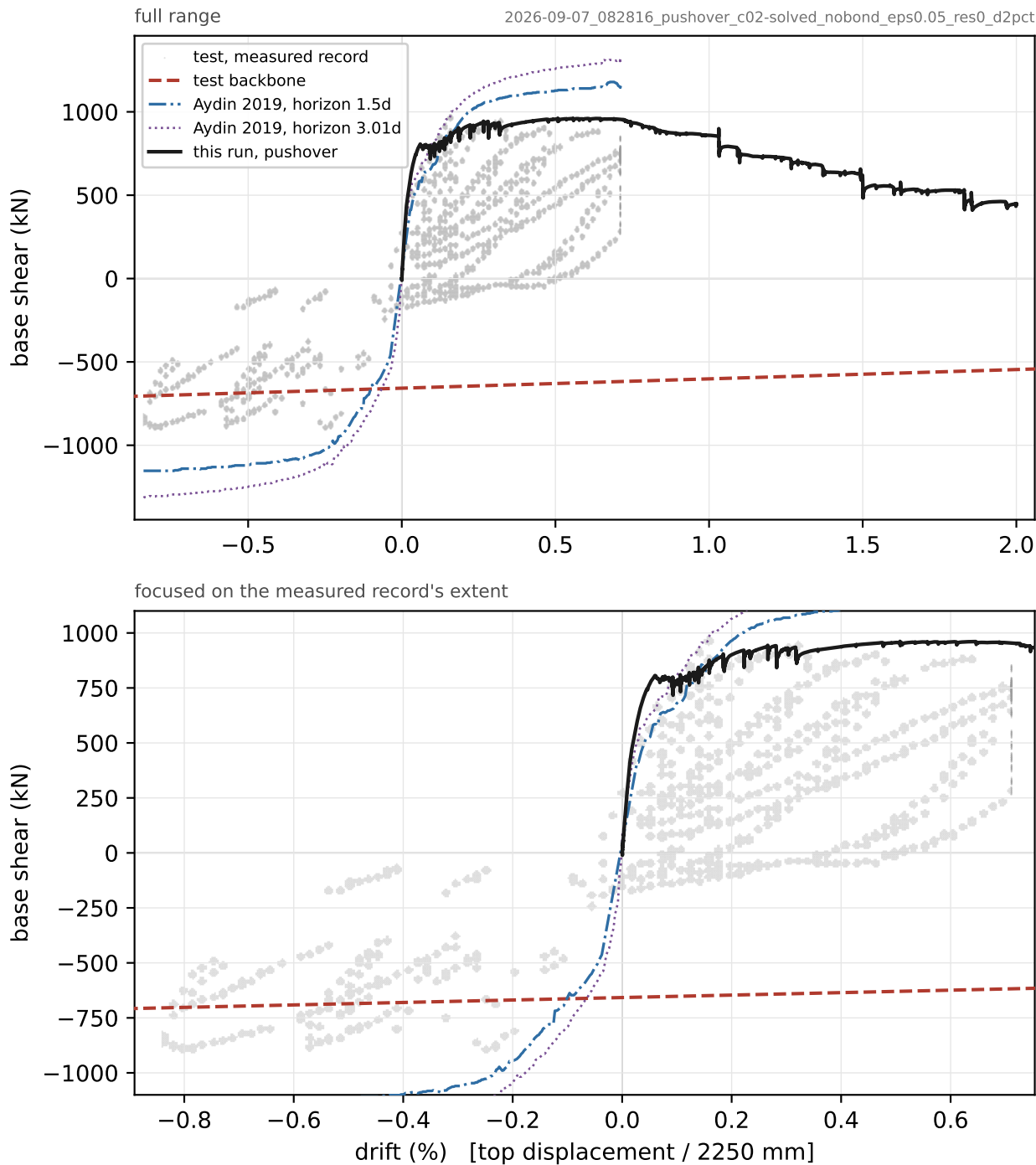


Figure 30: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.18 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — to 3% drift

2026-09-07_164126_pushover_c02-solved_nobond_eps0.05_d3pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$

bar rupture	0.05 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0245731, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:-5.6, epsU:-0.0180353, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,110,173
wall clock	2.03 h
converged	True

Recorded response

peak base shear	962 kN (1 ms moving average)
raw sample maximum	962.1 kN
drift at peak	0.647%
drift capacity (20% drop)	1.374% — basis: monotonic
traced to	3% drift
inertia + damping in the reaction, ascending branch (p95)	6.1 kN

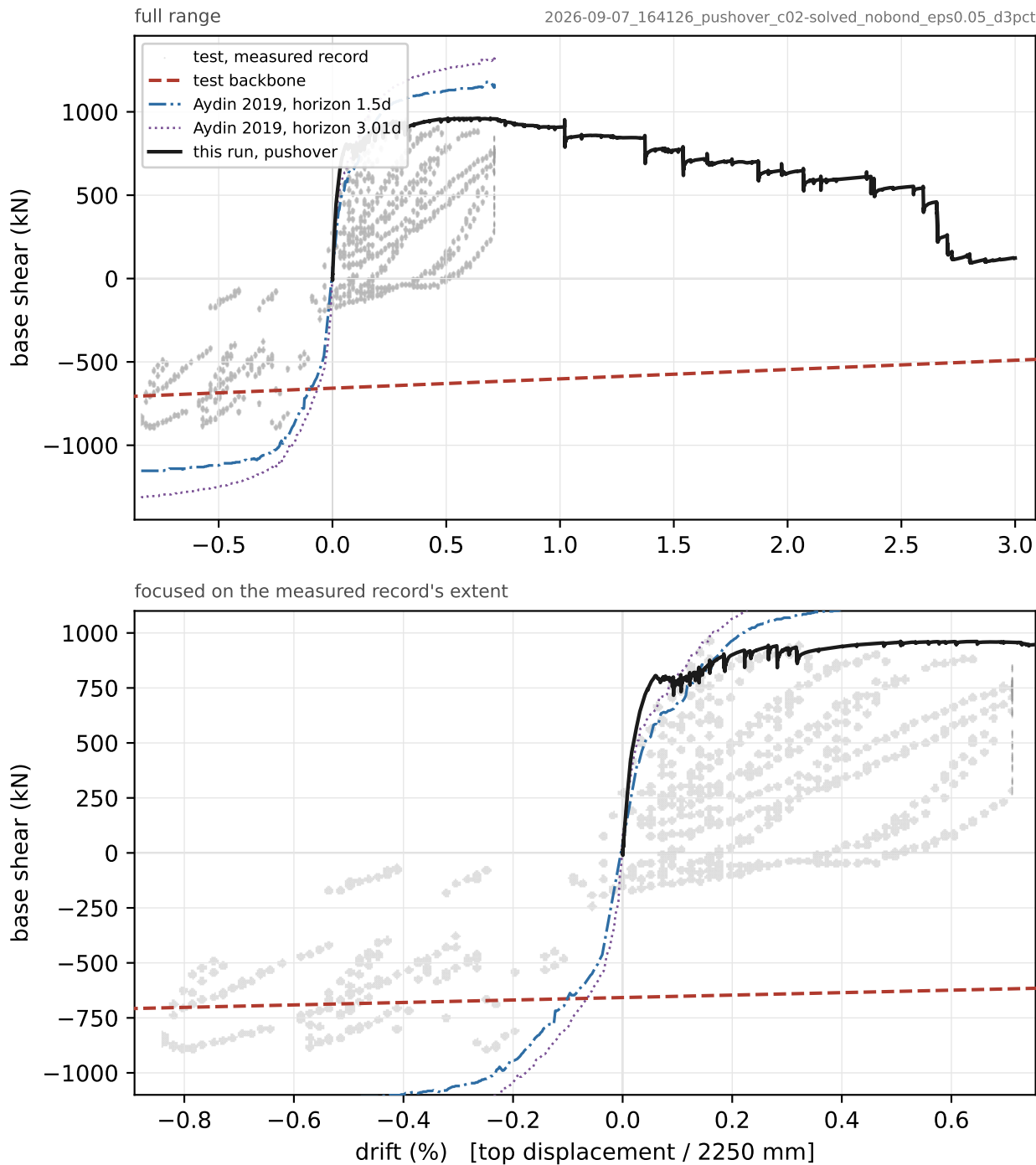


Figure 31: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

6.19 Pushover — crushing/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.075$ — residual 0 — to 3% drift

2026-09-07_204319_pushover_c02-solved_nobond_eps0.075_res0_d3pct

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1 (1.00 x Gf by construction (Concrete02, crack-band regularized per strut))

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	Concrete02 in compression: parabola to f_c , linear softening to f_{cu} , flat residual beyond
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$

bar rupture	0.075 strain
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	bond_damage, steel_b — this run predates them

tag	type	arguments	applied to
1	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0290374, lambda:0.1, ft:1.85, Ets:1140.83	5,505 concrete elements, L = 50 mm
2	Concrete02	fpc:-28, epsc0:-0.00225171, fpcu:0, epsU:-0.0211921, lambda:0.1, ft:1.85, Ets:1613.38	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.075, -max:0.075	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 us, 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	1,110,173
wall clock	3.38 h
converged	True

Recorded response

peak base shear	961 kN (1 ms moving average)
raw sample maximum	961.7 kN
drift at peak	0.646%
drift capacity (20% drop)	1.155% — basis: monotonic
traced to	3% drift
inertia + damping in the reaction, ascending branch (p95)	7.7 kN

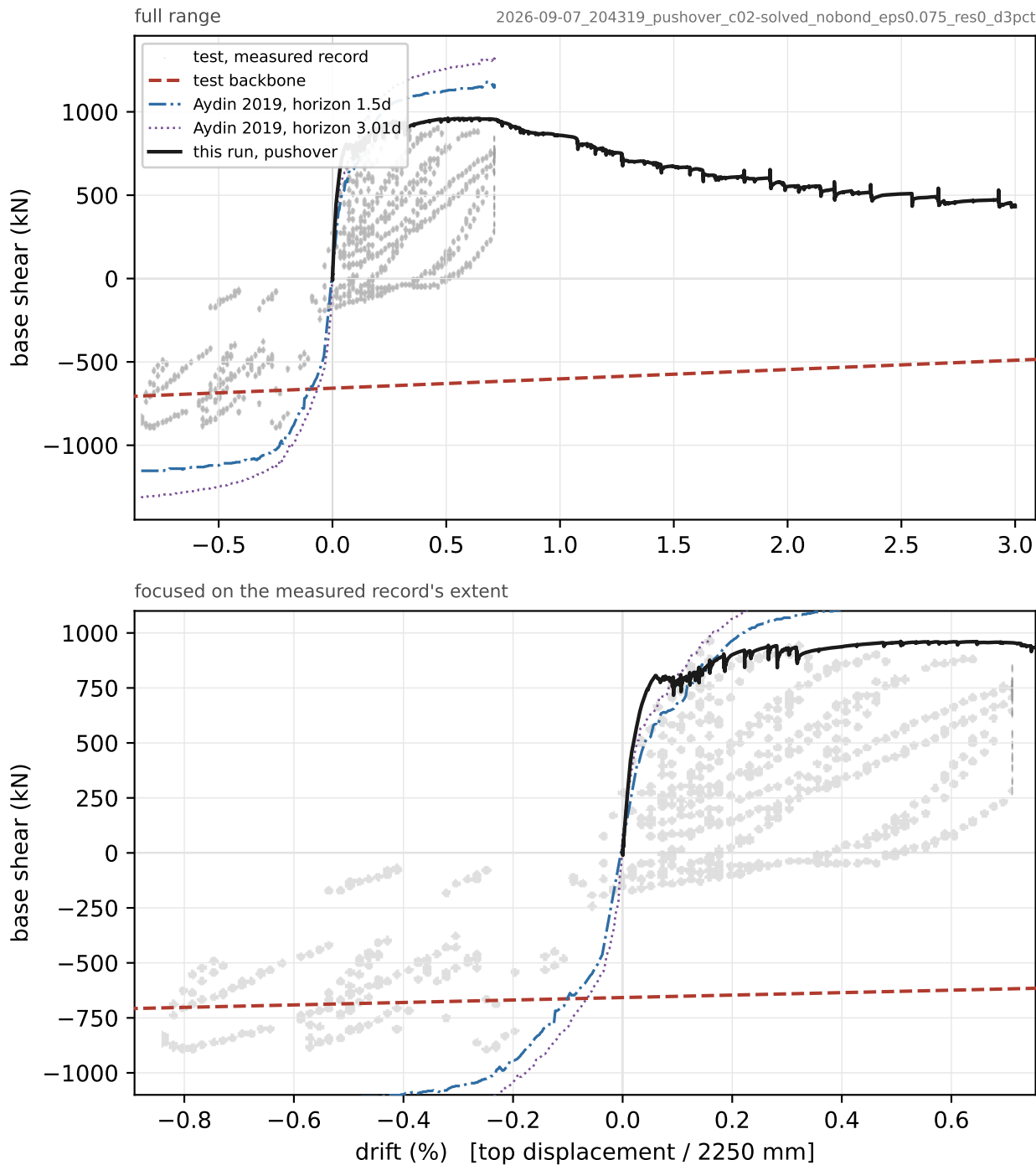


Figure 32: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

7 linear compression — paper tension tail — perfect bond

7.1 Pushover — linear/paper/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_200830_pushover_lincomp-paper_nobond

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times$ mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7$ mm ²
$E_t A_t$ per strut	$157.3e6$ N
G_f multiple of the tail	5.26 (5.26 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	linear elastic in compression, never yields
tension tail	the printed a_2, a_3 of Table 1, transplanted from the 20 mm grid
crushing residual	$0.2 \times f_c$

steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00520708, 0.0267793, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, $L = 50 \text{ mm}$
2	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00520708, 0.0267793, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, $L = 70.71 \text{ mm}$
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, $L = 50 \text{ mm}$

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	$L \text{ (mm)}$	$A \text{ (mm}^2\text{)}$	$E \text{ (MPa)}$	$EA \text{ (N)}$	$EA/L \text{ (N/mm)}$
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μs , 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	111,018
wall clock	0.22 h
converged	True

Recorded response

peak base shear	1,155.5 kN (1 ms moving average)
raw sample maximum	1,158 kN
drift at peak	0.284%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.3% drift
inertia + damping in the reaction,	6.7 kN
ascending branch (p95)	

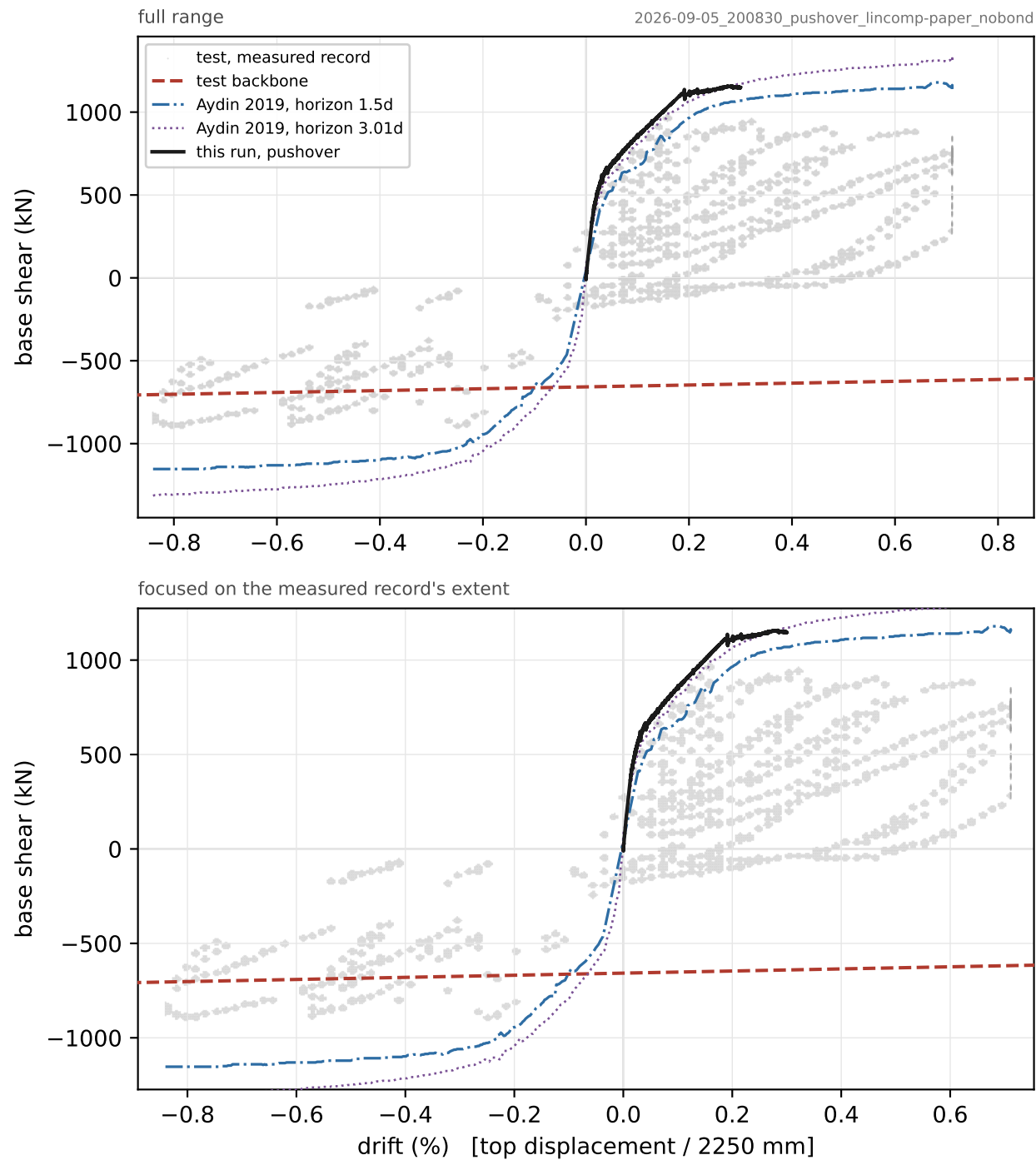


Figure 33: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

8 linear compression — solved tension tail — bond links, $0.6f_t$ residual

8.1 Pushover — linear/solved/bond60 — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_223007_pushover_lincomp-solved_bond-a06

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	5,484 / 34,775
composition	5,484 nodes, 34,775 elements (21,244 bond, 10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$

steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	bond links, residual plateau $0.6 f_t$

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, $L = 50 \text{ mm}$
2	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, $L = 70.71 \text{ mm}$
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, $L = 50 \text{ mm}$
4	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,652 bond elements, $L = 50 \text{ mm}$
5	ElasticMultiLinear	-strain:[-0.2, -7.44612e-05, -7.43868e-05, 0, 7.43868e-05, 7.44612e-05, 0.2], -stress:[-1.11, -1.11, -1.85, 0, 1.85, 1.11, 1.11]	10,592 bond elements, $L = 70.71 \text{ mm}$

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	$L \text{ (mm)}$	$A \text{ (mm}^2\text{)}$	$E \text{ (MPa)}$	$EA \text{ (N)}$	$EA/L \text{ (N/mm)}$
bond	10,652	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
bond	10,592	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	3.58 μs , 962 steps per fundamental period
first period T_1	3.444 ms

drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	248,089
wall clock	1.6 h
converged	True

Recorded response

peak base shear	1,286 kN (1 ms moving average)
raw sample maximum	1,373.4 kN
drift at peak	0.036%
drift capacity (20% drop)	0.038% — basis: monotonic
traced to	0.3% drift
inertia + damping in the reaction,	76 kN
ascending branch (p95)	

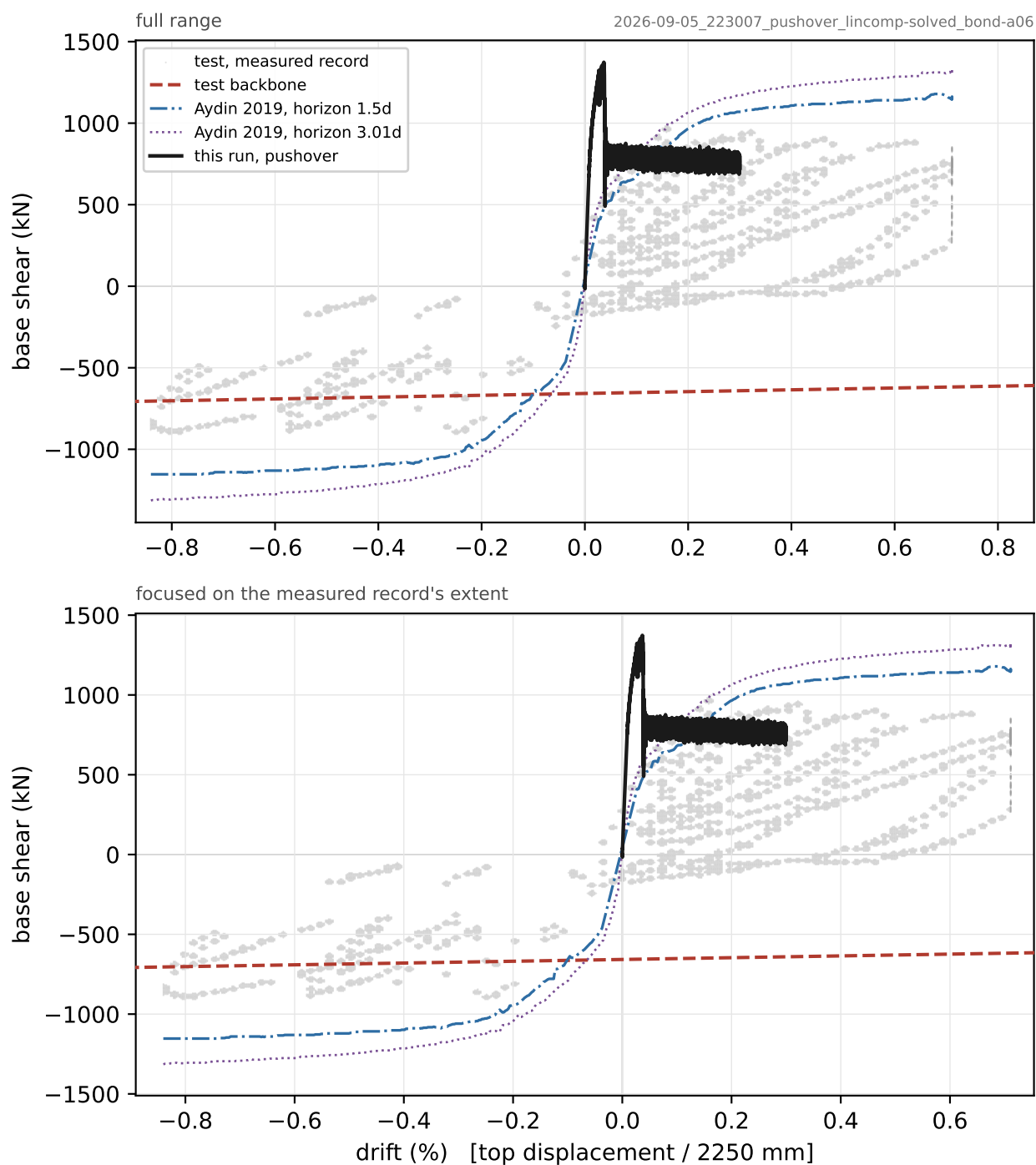


Figure 34: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

9 linear compression — solved tension tail — perfect bond

9.1 Pushover — linear/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 0.3% drift

2026-09-05_195507_pushover_lincomp-solved_nobond

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$

steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break
bond	perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance	rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used
parameters filled from today's defaults	<code>bond_damage</code> , <code>steel_rupture</code> , <code>concrete_residual</code> , <code>steel_b</code> — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, $L = 50 \text{ mm}$
2	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, $L = 70.71 \text{ mm}$
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, $L = 50 \text{ mm}$

Strut families — axial stiffness

note	E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record
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family	n	element	$L \text{ (mm)}$	$A \text{ (mm}^2\text{)}$	$E \text{ (MPa)}$	$EA \text{ (N)}$	$EA/L \text{ (N/mm)}$
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis	pushover
target	0.3% drift
solution scheme	dynamic relaxation, quasi-static
integrator	CentralDifference
time step	8 μs , 728 steps per fundamental period
first period T_1	5.824 ms
drive rate	7.6 mm/s
damping	0.5 critical, mass proportional
steps	111,018
wall clock	0.2 h
converged	True

Recorded response

peak base shear	941.6 kN (1 ms moving average)
raw sample maximum	944.2 kN
drift at peak	0.3%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	0.3% drift
inertia + damping in the reaction, ascending branch (p95)	12.1 kN

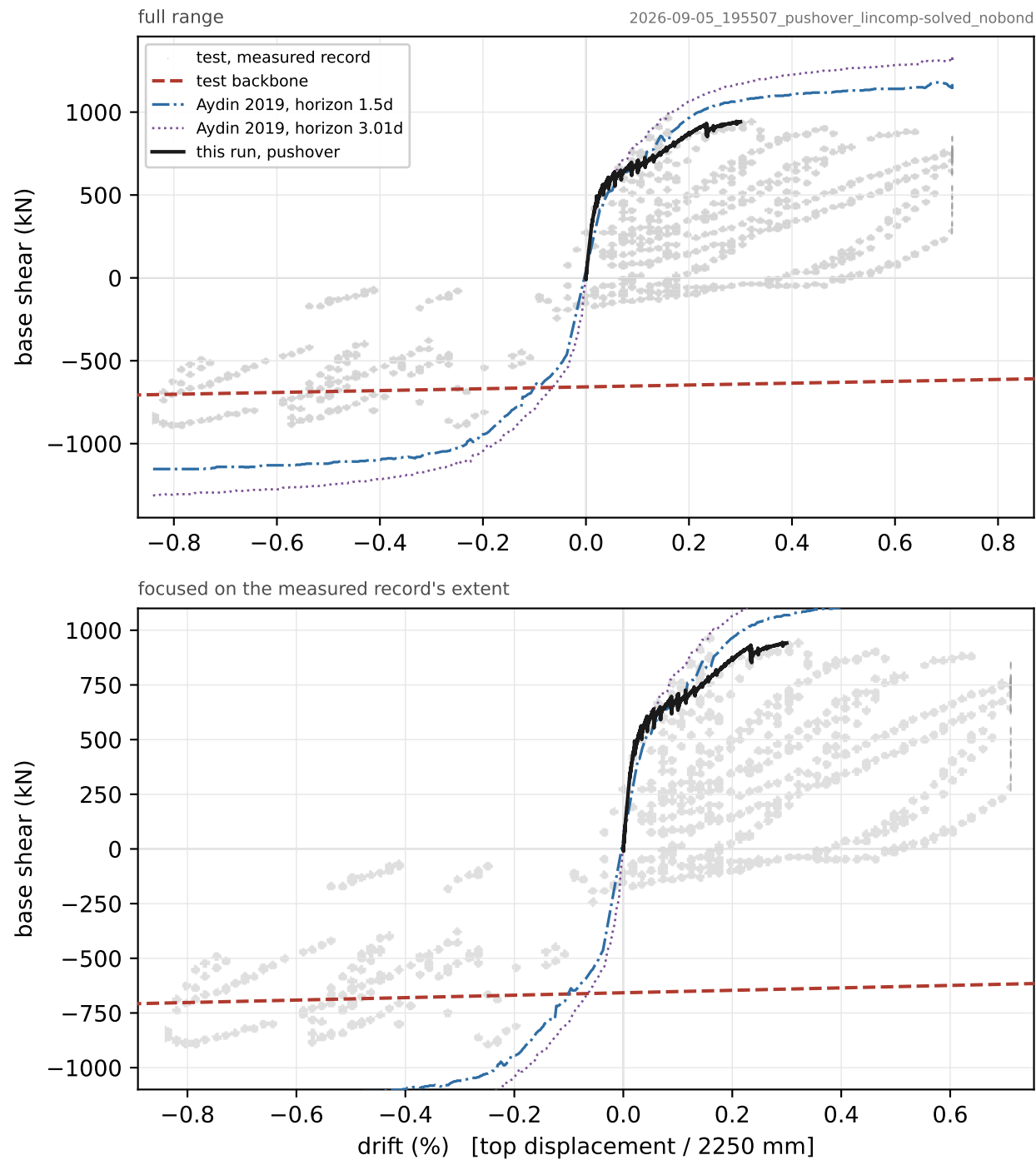


Figure 35: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

9.2 Pushover — linear/solved/perfect — mesh 50 mm — energy balance, uniaxial field — to 2% drift

2026-09-06_000826_pushover_lincomp-solved_nobond_d2pct

Specimen and discretisation

panel ($L \times H \times t$)	$3,000 \times 2,250 \times 210$ mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	$1.5 \times \text{mesh}$
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	$6,324.7 \text{ mm}^2$
$E_t A_t$ per strut	157.3 e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28 \text{ MPa}$, $f_t = 1.85 \text{ MPa}$, $G_f = 0.075 \text{ N/mm}$
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360 \text{ MPa}$, $E_s = 200,000 \text{ MPa}$, $b = 0.01$
bar rupture	none — bars cannot break

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults `bond_damage`, `steel_rupture`, `concrete_residual`, `steel_b` — this run pre-dates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, L = 50 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
concrete, orthogonal	5,505	corotTruss	50	6,324.7	24,870	1.573e8	3.146e6
concrete, diagonal	5,400	corotTruss	70.71	6,324.7	24,870	1.573e8	2.225e6
longitudinal	1,350	corotTruss	50	150.8	200,000	3.016e7	6.032e5
stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis pushover

target 2% drift

solution scheme dynamic relaxation, quasi-static

integrator CentralDifference

time step 8 us, 728 steps per fundamental period

first period T_1 5.824 ms

drive rate 7.6 mm/s

damping 0.5 critical, mass proportional

steps 740,116

wall clock 1.32 h

converged True

Recorded response

peak base shear 1,052.3 kN (1 ms moving average)

raw sample maximum 1,054.6 kN

drift at peak	1.999%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	2% drift
inertia + damping in the reaction,	4.1 kN
ascending branch (p95)	

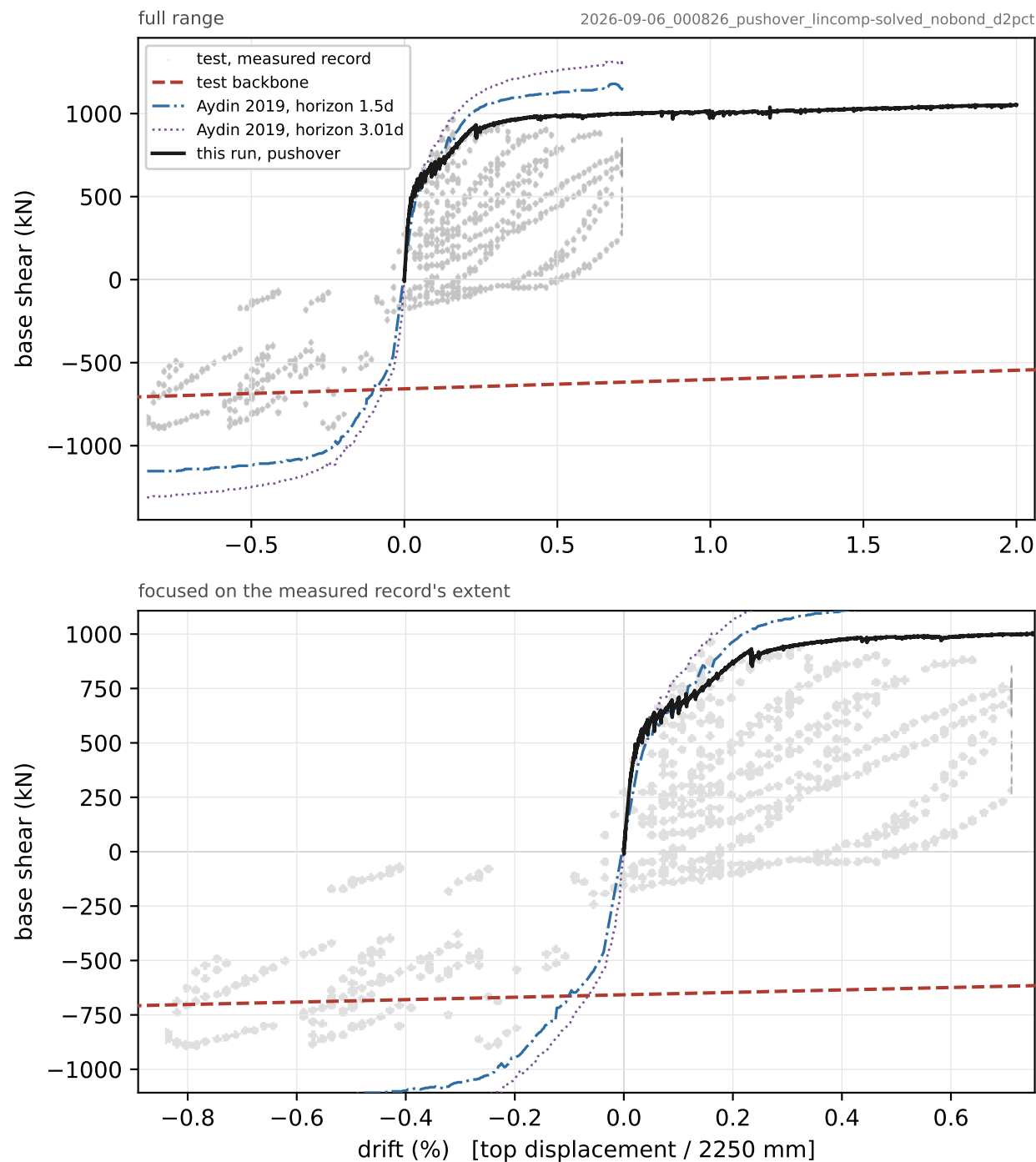


Figure 36: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.

9.3 Pushover — linear/solved/perfect — mesh 50 mm — energy balance, uniaxial field — $\varepsilon_{su} = 0.05$ — to 3% drift

2026-09-09_000702_pushover_linear-solved_perfect_eps0.05_d3pct

Specimen and discretisation

panel ($L \times H \times t$)	3,000 × 2,250 × 210 mm
drift denominator	2,250 mm
mesh size	50 mm
strut horizon	1.5 × mesh
nodes / elements	2,806 / 13,531
composition	2,806 nodes, 13,531 elements (10,905 concrete, 1,350 longitudinal, 1,276 stirrup)
reinforcement	included, bars run to the top face
strut life $\varepsilon_u/\varepsilon_{cr}$	orthogonal 22.8, diagonal 16.4
first cracking	259 kN

Calibration

strategy	Aydin's overlapping-lattice energy balance: the stored elastic energy of the continuum and of the lattice are equated under an affine strain field and solved for one uniform $E_t A_t$. A closed-form sum over the strut list — no FE solve, no reference model, no optimiser.
affine field	uniaxial
Poisson ratio ν in the balance	0.2
strut area A_t	6,324.7 mm ²
$E_t A_t$ per strut	157.3e6 N
G_f multiple of the tail	1.05 (1.05 x G_f per cracking strut at $L = 50.0$ mm)

Under an affine displacement field $\mathbf{u} = \mathbf{F}\mathbf{x}$ a strut spanning $\mathbf{v}_s = \mathbf{x}_b - \mathbf{x}_a$, of length $L_s = \|\mathbf{v}_s\|$, carries the axial strain

$$\varepsilon_s = \frac{(\mathbf{F}\mathbf{v}_s) \cdot \mathbf{v}_s}{L_s^2}, \quad W_{\text{lattice}} = \frac{E_t A_t}{2} \sum_s \varepsilon_s^2 L_s.$$

The continuum energy is taken under uniaxial strain with the transverse strain restrained ($\varepsilon_x = \varepsilon$, $\varepsilon_y = 0$), plane stress:

$$W_{\text{continuum}} = \frac{E \varepsilon^2 V}{2(1 - \nu^2)}, \quad V = t A.$$

Equating the two and solving for the one unknown,

$$E_t A_t = \frac{W_{\text{continuum}}}{\frac{1}{2} \sum_s \varepsilon_s^2 L_s}, \quad A_t = \frac{E_t A_t}{E}.$$

Both sides are quadratic in ε , so the strain magnitude cancels exactly and the result is a property of the grid, the horizon, ν and the thickness alone — it transfers to any model built on the same grid. For an interior node the equibiaxial balance collapses to the published closed form $E_t A_t = C E_t d w$ with $C = 0.621$ at a $1.5d$ horizon and $C = 0.102$ at $3.01d$ (2019 Appendix, Eq 6, at $\nu = 1/3$).

Constitutive inputs

concrete	$f_c = 28$ MPa, $f_t = 1.85$ MPa, $G_f = 0.075$ N/mm
compression branch	linear elastic in compression, never yields
tension tail	a_2, a_3 re-solved at this mesh so the tail dissipates G_f per strut
crushing residual	$0.2 \times f_c$
steel	$f_y = 360$ MPa, $E_s = 200,000$ MPa, $b = 0.01$
bar rupture	0.05 strain

bond perfect bond: rebar shares the concrete nodes

OpenSees materials

provenance rebuilt from this run's own parameters; the build is deterministic, so these are the records the run used

parameters filled from today's defaults `steel_b` — this run predates them

tag	type	arguments	applied to
1	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.00101436, 0.0052138, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,505 concrete elements, L = 50 mm
2	ElasticMultiLinear	-strain:[-0.2, -0.00075057, 0, 7.43868e-05, 0.00011158, 0.000722613, 0.00371423, 0.2], -stress:[-2000.8, -18.6667, 0, 1.85, 1.11, 0.37, 0, 0]	5,400 concrete elements, L = 70.71 mm
3	Steel02	Fy:360, E0:200000, b:0.01, R0:18, cR1:0.925, cR2:0.15	referenced by tag 4
4	MinMax	otherTag:3, -min:-0.05, -max:0.05	2,626 longitudinal/stirrup elements, L = 50 mm

Strut families — axial stiffness

note E is the material's INITIAL axial modulus; for Concrete02 that is $2f_c/\varepsilon_{c0}$, which is the modulus the calibration was struck against, not an argument of the record

family	n	element	L (mm)	A (mm ²)	E (MPa)	EA (N)	EA/L (N/mm)
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stirrup	1,276	corotTruss	50	150.8	200,000	3.016e7	6.032e5

Analysis and solver

analysis pushover

target 3% drift

solution scheme dynamic relaxation, quasi-static

integrator CentralDifference

time step 8 us, 728 steps per fundamental period

first period T_1 5.824 ms

drive rate 7.6 mm/s

damping 0.5 critical, mass proportional

steps 1,110,173

wall clock 2.05 h

converged True

Recorded response

peak base shear	1,079.3 kN (1 ms moving average)
raw sample maximum	1,118.3 kN
drift at peak	2.989%
drift capacity (20% drop)	not reached — basis: monotonic
traced to	3% drift
inertia + damping in the reaction,	4 kN
ascending branch (p95)	

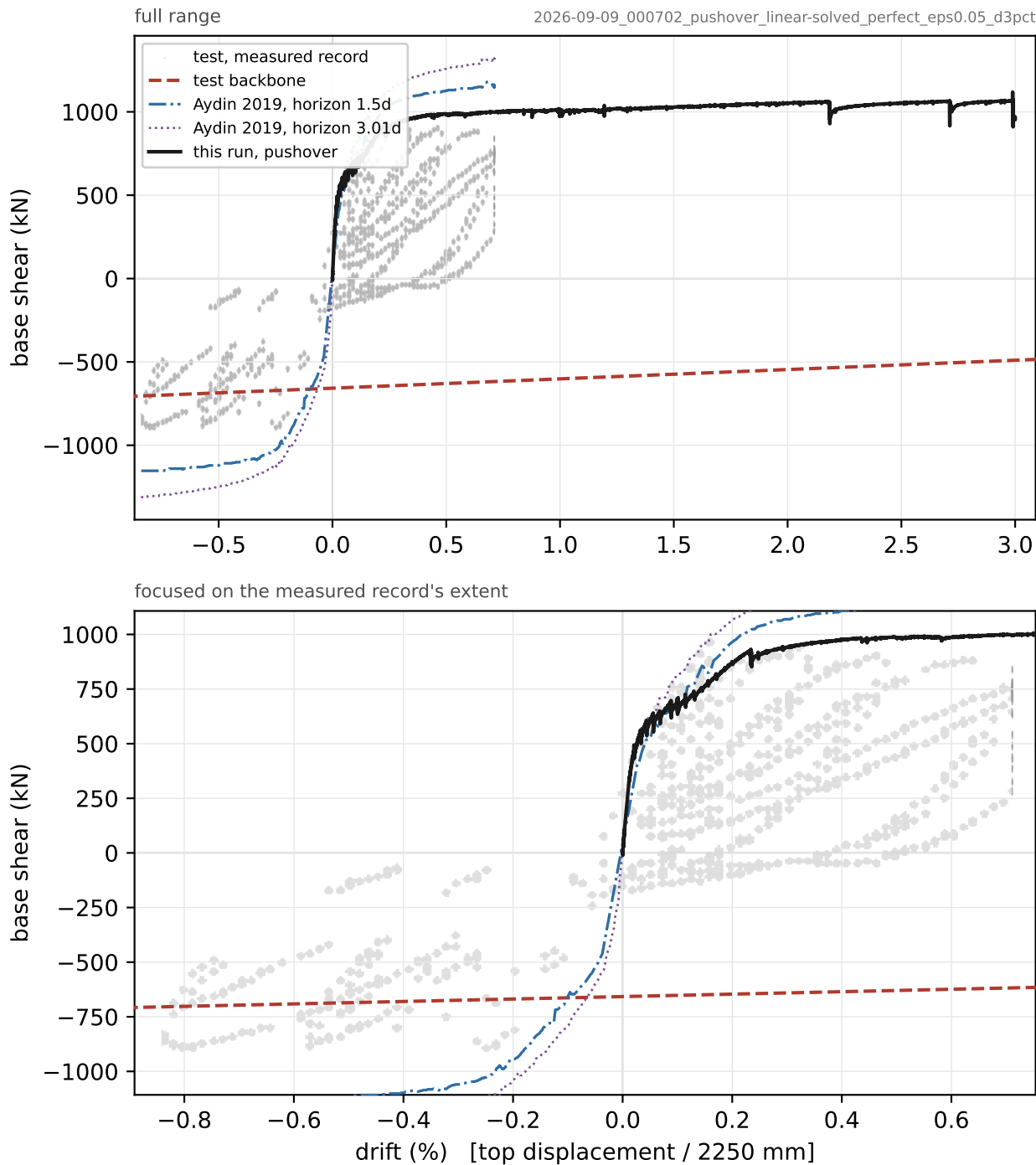


Figure 37: Pushover over the measured cyclic record, its envelope, and the two simulated curves of Aydin et al. (2019); full range above, the record's own extent below.